

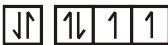
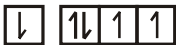
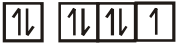
INORGANIC CHEMISTRY

E

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ATOMIC STRUCTURE

EXERCISE-I

- The following quantum no. are possible for how many orbitals $n = 3$, $\ell = 2$, $m = +2$
(1) 1 (2) 2
(3) 3 (4) 4
- The energy of second Bohr orbit of the hydrogen atom is -328 KJ/mol . Hence the energy of fourth Bohr orbit should be :
(1) -41 KJ/mol (2) -1312 KJ/mol
(3) -164 KJ/mol (4) -82 KJ/mol
- The measurement of the electron position is associated with an uncertainty in momentum, which is equal to $1 \times 10^{-18} \text{ g cm s}^{-1}$. the uncertainty in electron velocity is : (mass of electron = $9 \times 10^{-28} \text{ g}$)
(1) $1 \times 10^{11} \text{ cm s}^{-1}$
(2) $1 \times 10^9 \text{ cm s}^{-1}$
(3) $1 \times 10^6 \text{ cm s}^{-1}$
(4) $1 \times 10^5 \text{ cm s}^{-1}$
- Maximum number of electrons in a subshell of an atom is determined by the following :-
(1) $2n^2$ (2) $4\ell + 2$
(3) $2\ell + 1$ (4) $4\ell - 2$
- A 0.66 kg ball is moving with a speed of 100 m/s . The associated wavelength will be
($h = 6.6 \times 10^{-34} \text{ Js}$) :-
(1) $6.6 \times 10^{-34} \text{ m}$ (2) $1.0 \times 10^{-35} \text{ m}$
(3) $1.0 \times 10^{-32} \text{ m}$ (4) $6.6 \times 10^{-32} \text{ m}$
- The energies E_1 and E_2 of two radiations are 25 eV and 50 eV respectively. The relation between their wavelengths i.e. λ_1 and λ_2 will be :
(1) $\lambda_1 = \lambda_2$ (2) $\lambda_1 = 2\lambda_2$
(3) $\lambda_1 = 4\lambda_2$ (4) $\lambda_1 = \frac{1}{2}\lambda_2$
- Smallest wavelength occurs for
(1) Lyman series
(2) Balmer series
(3) Paschen series
(4) Brackett series
- Maximum number of electrons in a subshell with $\ell = 3$ and $n = 4$ is:
(1) 10 (2) 12
(3) 14 (4) 16
- The value of Planck's constant is $6.63 \times 10^{-34} \text{ Js}$. The speed of light is $3 \times 10^{17} \text{ nm s}^{-1}$. Which value is closest to the wavelength in nanometer of a quantum of light with frequency of $6 \times 10^{15} \text{ s}^{-1}$?
(1) 75 (2) 10
(3) 25 (4) 50
- The energy of an electron of $2p_y$ orbital is
(1) greater than $2p_x$ orbital
(2) Less than $2p_z$ orbital
(3) same as that of $2p_x$ and $2p_z$ orbital
(4) Equal to $2s$ orbital
- A and B are two elements which have same atomic weight and are having atomic number 27 and 30 respectively. If the atomic weight of A is 57 then number of neutron in B is :-
(1) 27 (2) 33 (3) 30 (4) 40
- Energy required to remove an e^- from M shell of H-atom is 1.51 eV , then energy of I^{st} excited state will be :-
(1) -1.51 eV (2) $+1.51 \text{ eV}$
(3) -3.4 eV (4) -13.6 eV
- Number of possible orbitals (all types) in $n = 3$ energy level is :-
(1) 1 (2) 3
(3) 4 (4) 9
- When $3d$ orbital is complete, the new electron enters into :-
(1) $4p$ orbital
(2) $4f$ orbital
(3) $4s$ orbital
(4) $4d$ orbital
- Which orbital diagram does not obey Aufbau principle :-
(1)  (2) 
(3)  (4) 
- Species which are isoelectronic to one another are
(a) CN^- (b) OH^-
(c) CH_3^+ (d) N_2
(e) CO
Correct Ans :-
(1) a, b, c (2) a, c, d
(3) a, d, e (4) b, c, d
- $\frac{h}{2\pi}$ is angular momentum in.....orbit of He^+
(1) First (2) Second
(3) Third (4) Infinite

18. 1st shell energy of He^+ is -54.4 eV . Then energy of its 2nd shell is :-
 (1) -54.4 eV (2) -13.6 eV
 (3) -27.2 eV (4) $+27.2 \text{ eV}$
19. Third line of Balmer series is produced by which transition in spectrum of H-atom
 (1) 5 to 2
 (2) 5 to 1
 (3) 4 to 2
 (4) 4 to 1
20. The ratio of radii of 3rd and 2nd Bohr's orbits of hydrogen atom is :-
 (1) 3 : 2 (2) 4 : 9
 (3) 9 : 4 (4) 9 : 1
21. A metal in its dipositive state has the electronic configuration 2, 8, 14 and has the atomic weight equal to 56. Number of neutrons in its nucleus would be
 (1) 30 (2) 32 (3) 34 (4) 28
22. In Balmer series of hydrogen atom spectrum which electronic transition causes third line :-
 (1) Fifth Bohr orbit to second
 (2) Fifth Bohr orbit to first
 (3) Fourth Bohr orbit to second
 (4) Fourth Bohr orbit to first
23. The ratio between kinetic energy and the total energy of the electrons of hydrogen atom according to Bohr's model is :-
 (1) 2 : 1 (2) 1 : 1
 (3) 1 : -1 (4) 1 : 2
24. Correct statement is :-
 (1) $\text{K} = 4s^1$, $\text{Cr} = 3d^4 4s^2$, $\text{Cu} = 3d^{10} 4s^2$
 (2) $\text{K} = 4s^2$, $\text{Cr} = 3d^4 4s^2$, $\text{Cu} = 3d^{10} 4s^2$
 (3) $\text{K} = 4s^2$, $\text{Cr} = 3d^5 4s^1$, $\text{Cu} = 3d^{10} 4s^2$
 (4) $\text{K} = 4s^1$, $\text{Cr} = 3d^5 4s^1$, $\text{Cu} = 3d^{10} 4s^1$
25. Which of the following pairs is correctly matched
 (1) Isotopes $^{40}_{20}\text{Ca}$, $^{40}_{19}\text{K}$
 (2) Isotones $^{30}_{14}\text{Si}$, $^{31}_{15}\text{P}$, $^{32}_{16}\text{S}$
 (3) Isobars $^{16}_8\text{O}$, $^{17}_8\text{O}$, $^{18}_8\text{O}$
 (4) Isolectronic N^{-3} , O^{-2} , Cr^{+3}
26. The relative abundance of two rubidium isotopes of atomic weights 85 and 87 are 75% and 25% respectively. The average atomic wt. of rubidium is:-
 (1) 75.5 (2) 85.5
 (3) 86.5 (4) 87.5
27. The ratio of specific charge of a proton and an α -particle is :-
 (1) 2 : 1 (2) 1 : 2
 (3) 1 : 4 (4) 1 : 1
28. In an atom $^{27}_{13}\text{Al}$, number of protons is (a) electron is (b) and neutron is (c). Hence ratio will be [in order c : b : a]
 (1) 13 : 14 : 13 (2) 13 : 13 : 14
 (3) 14 : 13 : 13 (4) 14 : 13 : 14
29. An isotone of $^{76}_{32}\text{Ge}$ is :-
 (i) $^{77}_{32}\text{Ge}$ (ii) $^{77}_{33}\text{As}$
 (iii) $^{77}_{34}\text{Se}$ (iv) $^{78}_{34}\text{Se}$
 (1) (ii) & (iii) (2) (i) & (ii)
 (3) (ii) & (iv) (4) (ii) & (iii) & (iv)
30. For Li^{+2} , $r_2 : r_5$ will be :-
 (1) 9 : 25 (2) 4 : 25
 (3) 25 : 4 (4) 25 : 9

				ANSWER KEY			Exercise-I			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	4	2	2	2	2	1	3	4	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	4	1	2	3	1	2	1	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	1	3	4	2	2	1	3	3	2

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

- The energy of an electron in the first Bohr orbit of H atom is -13.6 eV. The possible energy value(s) of the excited state(s) for electrons in Bohr orbits of hydrogen is/are : **[JEE 1998]**
 (1) -3.4 eV (2) -4.2 eV
 (3) -6.8 eV (4) $+6.8$ eV
- The number of nodal planes in a p_x orbital is: **[JEE 2000]**
 (1) one (2) two
 (3) three (4) zero
- An atom has a mass of 0.02 kg and uncertainty in its velocity is 9.218×10^{-6} m/s then uncertainty in position is ($h = 6.626 \times 10^{-34}$ Js) **[AIEEE 2002]**
 (1) 2.86×10^{-28} m (2) 2.86×10^{-32} cm
 (3) 1.5×10^{-27} m (4) 3.9×10^{-10} m
- Energy of H-atom in the ground state is -13.6 eV, Hence energy in the second excited state is- **[AIEEE 2002]**
 (1) -6.8 eV (2) -3.4 eV
 (3) -1.51 eV (4) -4.3 eV
- Uncertainty in position of a particle of 25 g in space is 10^{-5} m. Hence uncertainty in velocity (ms^{-1}) is (Planck's constant $h = 6.6 \times 10^{-34}$ Js) **[AIEEE-2002]**
 (1) 2.1×10^{-28} (2) 2.1×10^{-34}
 (3) 0.5×10^{-34} (4) 5.0×10^{-24}
- The orbital angular momentum for an electron revolving in an orbit is given by $\sqrt{\ell(\ell+1)} \cdot \frac{h}{2\pi}$. This momentum for an s-electron will be given by **[AIEEE-2003]**
 (1) $\sqrt{2} \cdot \frac{h}{2\pi}$ (2) $1 \cdot \frac{h}{2\pi}$
 (3) zero (4) $\frac{h}{2\pi}$
- The number of d-electrons retained in Fe^{2+} (At. no. of Fe = 26) ion is : **[AIEEE-2003]**
 (1) 6 (2) 3 (3) 4 (4) 5
- The de Broglie wavelength of a tennis ball of mass 60 g moving with a velocity of 10 metres per second is approximately : **[AIEEE 2003]**
 (1) 10^{-25} metres (2) 10^{-33} metres
 (3) 10^{-31} metres (4) 10^{-16} metres
- Which of the following sets of quantum number is correct for an electron in 4f orbital ? **[AIEEE-2004]**
 (1) $n = 3, l = 2, m = -2, s = +1/2$
 (2) $n = 4, l = 4, m = -4, s = -1/2$
 (3) $n = 4, l = 3, m = +1, s = +1/2$
 (4) $n = 4, l = 3, m = +4, s = +1/2$
- Consider the ground state of Cr atom ($Z = 24$). The numbers of electrons with the azimuthal quantum numbers, $l = 1$ and 2 are, respectively **[AIEEE-2004]**
 (1) 16 and 5 (2) 12 and 5
 (3) 16 and 4 (4) 12 and 4
- The wavelength of the radiation emitted, when in a hydrogen atom electron falls from infinity to stationary state 1, would be **[AIEEE-2004]** (Rydberg constant = $1.097 \times 10^7 \text{ m}^{-1}$) :
 (1) 9.1×10^{-8} nm (2) 192 nm
 (3) 406 nm (4) 91 nm
- Which one of the following sets of ions represents the collection of isoelectronic species ? **[AIEEE-2004]**
 (1) $\text{Na}^+, \text{Mg}^{2+}, \text{Al}^{3+}, \text{Cl}^-$ (2) $\text{Na}^+, \text{Ca}^{2+}, \text{Sc}^{3+}, \text{F}^-$
 (3) $\text{K}^+, \text{Cl}^-, \text{Mg}^{2+}, \text{Sc}^{3+}$ (4) $\text{K}^+, \text{Ca}^{2+}, \text{Sc}^{3+}, \text{Cl}^-$
- The radius of which of the following orbit is same as that of the first Bohr's orbit of hydrogen atom? **[JEE 2004]**
 (1) $\text{He}^+ (n = 2)$ (2) $\text{Li}^{2+} (n = 2)$
 (3) $\text{Li}^{2+} (n = 3)$ (4) $\text{Be}^{3+} (n = 2)$
- In a multi-electron atom, which of the following orbitals described by the three quantum members will have the same energy in the absence of magnetic and electric fields ? **[AIEEE-2005]**
 (A) $n = 1, l = 0, m = 0$
 (B) $n = 2, l = 0, m = 0$
 (C) $n = 2, l = 1, m = 1$
 (D) $n = 3, l = 2, m = 1$
 (E) $n = 3, l = 2, m = 0$
 (1) (D) and (E) (2) (C) and (D)
 (3) (B) and (C) (4) (A) and (B)
- Of the following sets which one does not contain isoelectronic species ? **[AIEEE-2005]**
 (1) $\text{BO}_3^{3-}, \text{CO}_3^{2-}, \text{NO}_3^-$ (2) $\text{SO}_3^{2-}, \text{CO}_3^{2-}, \text{NO}_3^-$
 (3) $\text{CN}^-, \text{N}_2, \text{C}_2^{2-}$ (4) $\text{PO}_4^{3-}, \text{SO}_4^{2-}, \text{ClO}_4^-$

- 16.** Which of the following statements in relation to the hydrogen atom is correct ? [AIEEE-2005]
 (1) 3s, 3p and 3d orbitals all have the same energy
 (2) 3s and 3p orbitals are of lower energy than 3d orbitals
 (3) 3p orbital is lower in energy than 3d orbital
 (4) 3s orbitals is lower in energy than 3p orbital
- 17.** According to Bohr's theory angular momentum of electron in 5th shell is :- [AIEEE-2006]
 (1) $1.0 \text{ h}/\pi$ (2) $10 \text{ h}/\pi$
 (3) $2.5 \text{ h}/\pi$ (4) $25 \text{ h}/\pi$
- 18.** Uncertainty in the position of an electron (mass = $9.1 \times 10^{-31} \text{ Kg}$) moving with a velocity 300 ms^{-1} , accurate upto 0.001%, will be :- ($h = 6.63 \times 10^{-34} \text{ Js}$) [AIEEE-2006]
 (1) $5.76 \times 10^{-2} \text{ m}$ (2) $1.92 \times 10^{-2} \text{ m}$
 (3) $3.84 \times 10^{-2} \text{ m}$ (4) $19.2 \times 10^{-2} \text{ m}$
- 19.** Which of the following sets of quantum numbers represents the highest energy of an atom ? [AIEEE-2007]
 (1) $n = 3, l = 1, m = 1, s = +1/2$
 (2) $n = 3, l = 2, m = 1, s = +1/2$
 (3) $n = 4, l = 0, m = 0, s = +1/2$
 (4) $n = 3, l = 0, m = 0, s = +1/2$
- 20.** The ionization enthalpy of hydrogen atom is $1.312 \times 10^6 \text{ J mol}^{-1}$. The energy required to excite the electron in the atom from $n = 1$ to $n = 2$ is [AIEEE-2008]
 (1) $8.51 \times 10^5 \text{ J mol}^{-1}$ (2) $6.56 \times 10^5 \text{ J mol}^{-1}$
 (3) $7.56 \times 10^5 \text{ J mol}^{-1}$ (4) $9.84 \times 10^5 \text{ J mol}^{-1}$
- 21.** In an atom, an electron is moving with a speed of 600 m/s with an accuracy of 0.005%. Certainty with which the position of the electron can be located is ($h = 6.6 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$, mass of electron, $m_e = 9.1 \times 10^{-31} \text{ kg}$):- [AIEEE-2009]
 (1) $1.92 \times 10^{-3} \text{ m}$ (2) $3.84 \times 10^{-3} \text{ m}$
 (3) $1.52 \times 10^{-4} \text{ m}$ (4) $5.10 \times 10^{-3} \text{ m}$
- 22.** Calculate the wavelength (in nanometer) associated with a proton moving at $1.0 \times 10^3 \text{ ms}^{-1}$ (Mass of proton = $1.67 \times 10^{-27} \text{ kg}$ and $h = 6.63 \times 10^{-34} \text{ Js}$):- [AIEEE-2009]
 (1) 2.5 nm (2) 14.0 nm
 (3) 0.032 nm (4) 0.40 nm
- 23.** Ionisation energy of He^+ is $19.6 \times 10^{-18} \text{ J atom}^{-1}$. The energy of the first stationary state ($n = 1$) of Li^{2+} is:- [AIEEE-2010]
 (1) $8.82 \times 10^{-17} \text{ J atom}^{-1}$
 (2) $4.41 \times 10^{-16} \text{ J atom}^{-1}$
 (3) $-4.41 \times 10^{-17} \text{ J atom}^{-1}$
 (4) $-2.2 \times 10^{-15} \text{ J atom}^{-1}$
- 24.** The frequency of light emitted for the transition $n = 4$ to $n = 2$ of He^+ is equal to the transition in H atom corresponding to which of the following [AIEEE-2011]
 (1) $n = 3$ to $n = 1$ (2) $n = 2$ to $n = 1$
 (3) $n = 3$ to $n = 2$ (4) $n = 4$ to $n = 3$
- 25.** The electrons identified by quantum numbers n and ℓ :- [AIEEE-2012]
 (a) $n = 4, \ell = 1$ (b) $n = 4, \ell = 0$
 (c) $n = 3, \ell = 2$ (d) $n = 3, \ell = 1$
 Can be placed in order of increasing energy as
 (1) (a) < (c) < (b) < (d) (2) (c) < (d) < (b) < (a)
 (3) (d) < (b) < (c) < (a) (4) (b) < (d) < (a) < (c)
- 26.** The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a_0 is Bohr radius] [JEE 2012]
 (1) $\frac{h^2}{4\pi^2 m a_0^2}$ (2) $\frac{h^2}{16\pi^2 m a_0^2}$
 (3) $\frac{h^2}{32\pi^2 m a_0^2}$ (4) $\frac{h^2}{32\pi^2 m a_0^2}$

PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	1	3	1	3	1	2	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	4	1	2	1	3	2	2	4
Que.	21	22	23	24	25	26				
Ans.	1	4	3	2	3	3				

PERIODIC TABLE

EXERCISE-I

- The correct order of atomic or ionic size :
 (1) $N < Li < B$ (2) $Cl < Mg < Ca$
 (3) $Ca^{+2} < S^{-2} < Cl^{-}$ (4) $Na^{+} < Mg^{+2} < Cl^{-}$
- The ionization energy will be maximum for the process :
 (1) $Ba \rightarrow Ba^{+}$ (2) $Be \rightarrow Be^{+}$
 (3) $Cs \rightarrow Cs^{+}$ (4) $Li \rightarrow Li^{+}$
- Of the following elements, which possesses the highest electron affinity?
 (1) As (2) O (3) S (4) Se
- On the Pauling's electronegativity scale, which element is next to F ?
 (1) Cl (2) O (3) Br (4) Ne
- Calculate the bond length of C-X bond if C-C bond length is $1.54 \text{ \AA}$ and X-X bond length is $1.2 \text{ \AA}$ and electronegativities of C and X are 2.0 and 3.0 respectively.
 (1) $2.74 \text{ \AA}$ (2) $1.37 \text{ \AA}$
 (3) $1.46 \text{ \AA}$ (4) $1.28 \text{ \AA}$
- Which of the following element has highest metallic character ?

Element	IP
(1) P	17 eV
(2) Q	2 eV
(3) R	10 eV
(4) S	13 eV
- The ionic radius of Cr is minimum in which of the following compounds ?
 (1) CrF_3 (2) $CrCl_3$ (3) Cr_2O_3 (4) K_2CrO_4
- The IP_1, IP_2, IP_3, IP_4 and IP_5 of an element are 7.1, 14.3, 34.5, 46.8, 162.2 eV respectively. The element is likely to be:-
 (1) Na (2) Si (3) F (4) Ca
- $M(g) \longrightarrow M^{+}(g) + e^{-}$, $\Delta H = 100 \text{ eV}$
 $M(g) \longrightarrow M^{2+}(g) + 2e^{-}$, $\Delta H = 250 \text{ eV}$
 Which is incorrect statement ?
 (1) IE_1 of $M(g)$ is 100 eV
 (2) IE_1 of $M^{+}(g)$ is 150 eV
 (3) IE_2 of $M(g)$ is 250 eV
 (4) IE_2 of $M(g)$ is 150 eV
- Which of the following pairs of elements have almost similar atomic radii ?
 (1) Zr, Hf (2) Mo, W (3) Co, Ni (4) All

- In the given process which oxidation state is more stable ?
 $M_{(g)} \longrightarrow M_{(g)}^{+}$ $IE_1 = 7.9 \text{ eV}$
 $M_{(g)}^{+} \longrightarrow M_{(g)}^{+2}$ $IE_2 = 15.5 \text{ eV}$
 (1) M^{+} (2) M^{+2} (3) Both (4) None
- Which arrangement represents the correct order of electron gain enthalpy (with negative sign) of the given atomic species ?
 (1) $S < O < Cl < F$ (2) $O < S < F < Cl$
 (3) $Cl < F < S < O$ (4) $F < Cl < O < S$
- Electronegativity scale of Pauling is based upon :-
 (1) Bond length (2) Bond energy
 (3) Atomic radius (4) Covalent radius
- The element having electronic configuration $4f^{14} 5d^0 6s^2$ belongs to :-
 (1) d-block, 12th group
 (2) f-block, III B group
 (3) f-block, 14th group
 (4) s-block, 2nd group
- K^{+} , Ar , Ca^{2+} and S^{2-} contains -
 (1) Same electronic configuration and atomic volume
 (2) Different electronic configuration but same IP.
 (3) Same electronic configuration but different atomic volume
 (4) None
- The radius of potassium atom is 0.203 nm. The radius of the potassium ion in nanometer will be :-
 (1) 0.133 (2) 0.231
 (3) 0.234 (4) 0.251
- The covalent and vander Waal's radii of hydrogen respectively are :-
 (1) $0.37 \text{ \AA}$, $0.8 \text{ \AA}$ (2) $0.37 \text{ \AA}$, $0.37 \text{ \AA}$
 (3) $0.8 \text{ \AA}$, $0.8 \text{ \AA}$ (4) $0.8 \text{ \AA}$, $0.37 \text{ \AA}$
- The element having highest I.P. from the two series C, N, O and Si, P, S :-
 (1) P (2) N (3) S (4) O
- In which of the following pairs, the ionisation energy of the first species is less than that of the second ?
 (1) N, P (2) Be^{+} , Be
 (3) N, N^{-} (4) Ne, Ne^{+}

20. IE_1 , IE_2 and IE_3 of an element are 10 eV, 15 eV, 45 eV respectively, the most stable oxidation state of the element will be :-
 (1) + 1 (2) + 2 (3) + 3 (4) + 4
21. Which of the following is energy releasing process?
 (1) $X^- \rightarrow X(g) + e^-$
 (2) $O^-(g) + e^- \rightarrow O^{2-}$
 (3) $O(g) \rightarrow O^+(g) + e^-$
 (4) $O(g) + e^- \rightarrow O^-(g)$
22. Among the following least and most polar bonds are respectively :-
 (a) C - I (b) N - O (c) C - F (d) P - F
 (1) d and c (2) a and d
 (3) b and d (4) b and c
23. Which compound strongly absorb CO_2 ?
 (1) BeO (2) K_2O
 (3) H_3PO_4 (4) P_4O_6
24. Which of the following is different from other three oxides?
 (1) MgO (2) SnO (3) PbO (4) ZnO
25. ${}_{92}U$ (IIIB) changes to ${}_{90}Th$ by emission of α -particle. Daughter element will be in -
 (1) IB (2) IIA (3) IIIB (4) VB
26. The IE_1 & IE_2 of three elements A, B & C are given as (IE in KJ/mol).
- | | A | B | C |
|--------|------|------|------|
| IE_1 | 400 | 550 | 1150 |
| IE_2 | 2650 | 1070 | 2090 |
- Identify the element which represent a non-metal:-
 (1) A (2) B
 (3) Both A & B (4) C
27. The pair of amphoteric hydroxide is
 (1) $Al(OH)_3$, LiOH
 (2) $Be(OH)_2$, $Mg(OH)_2$
 (3) $B(OH)_3$, $Be(OH)_2$
 (4) $Be(OH)_2$, $Zn(OH)_2$
28. Which process requires maximum energy?
 (1) $Na(g) \rightarrow Na^+(g) + e^-$
 (2) $Al^{+3}(g) \rightarrow Al^{+4}(g) + e^-$
 (3) $Al^{+2}(g) \rightarrow Al^{+3}(g) + e^-$
 (4) $Na^+(g) \rightarrow Na^{+2}(g) + e^-$
29. Highest electron affinity observe in :-
 (1) $2s^2 2p^5$ (2) $2s^2 2p^4$
 (3) $2s^2 2p^3$ (4) $2s^2 2p^1$
30. Ionization potential of Na equals to the :-
 (1) Electron affinity of Na^+
 (2) Electronegativity of Na^+
 (3) Electron affinity of He
 (4) Ionization potential of Mg

				ANSWER KEY			Exercise-I			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	3	2	4	2	4	2	3	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	2	2	3	1	1	2	4	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	2	2	1	3	4	4	2	1	1

EXERCISE-II

1. Moving from right to left in a periodic table, the atomic size is : **[JEE 1995]**
 (1) increased (2) decreased
 (3) remains constant (4) none of these
 2. The increasing order of electronegativity in the following elements : **[JEE 1995]**
 (1) C, N, Si, P (2) N, Si, C, P
 (3) Si, P, C, N (4) P, Si, N, C
 3. Which of the following has the maximum number of unpaired electrons ? **[JEE 1996]**
 (1) Mg^{2+} (2) Ti^{3+} (3) V^{3+} (4) Fe^{2+}
 4. The correct order of radii is : **[JEE 2000]**
 (1) $\text{N} < \text{Be} < \text{B}$ (2) $\text{F}^- < \text{O}^{2-} < \text{N}^{3-}$
 (3) $\text{Na} < \text{Li} < \text{K}$ (4) $\text{Fe}^{3+} < \text{Fe}^{2+} < \text{Fe}^{4+}$
 5. Which of the following groupings represent a collection of isoelectronic species ? **[AIIEEE-2003]**
 (At. no. Cs = 55, Br = 35)
 (1) N^{3-} , F^- , Na^+ (2) Be , Al^{3+} , Cl^-
 (3) Ca^{2+} , Cs^+ , Br (4) Na^+ , Ca^{2+} , Mg^{2+}
 6. According to the Periodic law of elements, the variation in properties of elements is related to their :- **[AIIEEE-2003]**
 (1) Nuclear masses
 (2) Atomic numbers
 (3) Nuclear neutron-proton number ratio
 (4) Atomic masses
 7. Among Al_2O_3 , SiO_2 , P_2O_5 and SO_3 , the correct order of acid strength is :- **[AIIEEE-2004]**
 (1) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{SO}_3 < \text{P}_2\text{O}_5$
 (2) $\text{SiO}_2 < \text{SO}_3 < \text{Al}_2\text{O}_3 < \text{P}_2\text{O}_5$
 (3) $\text{SO}_3 < \text{P}_2\text{O}_5 < \text{SiO}_2 < \text{Al}_2\text{O}_3$
 (4) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{P}_2\text{O}_5 < \text{SO}_3$
 8. The formation of the oxide ion $\text{O}^{2-}(\text{g})$ requires first an exothermic and then an endothermic step as shown below :- **[AIIEEE-2004]**

$$\text{O}(\text{g}) + \text{e}^- = \text{O}^-(\text{g}), \Delta H^\circ = -142 \text{ kJ mol}^{-1}$$

$$\text{O}^-(\text{g}) + \text{e}^- = \text{O}^{2-}(\text{g}), \Delta H^\circ = 844 \text{ kJ mol}^{-1}$$
 This is because :-
 (1) O^- ion will tend to resist the addition of another electron
 (2) Oxygen has high electron affinity
 (3) Oxygen is more electronegative
 (4) O^- ion has comparatively larger size than oxygen atom
 9. Of the following sets which one does not contain isoelectronic species ? **[AIIEEE-2005]**
 (1) BO_3^{3-} , CO_3^{2-} , NO_3^- (2) SO_3^{2-} , CO_3^{2-} , NO_3^-
 (3) CN^- , N_2 , C_2^{2-} (4) PO_4^{3-} , SO_4^{2-} , ClO_4^-
 10. In which of the following arrangements the order is NOT according to the property indicated below? **[AIIEEE-2005]**
 (1) $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+ < \text{F}^-$ – increasing ionic size
 (2) $\text{B} < \text{C} < \text{N} < \text{O}$ – increasing first ionization enthalpy
 (3) $\text{I} < \text{Br} < \text{F} < \text{Cl}$ – increasing electron gain enthalpy (with negative sign)
 (4) $\text{Li} < \text{Na} < \text{K} < \text{Rb}$ – increasing metallic radius
 11. Which of the following oxides is amphoteric in character ? **[AIIEEE-2005]**
 (1) SnO_2 (2) SiO_2 (3) CO_2 (4) CaO
 12. Which of the following factors may be regarded as the main cause of lanthanide contraction ? **[AIIEEE-2005]**
 (1) poor shielding of one of 4f electron by another in the subshell
 (2) effective shielding of one of 4f electrons by another in the subshell
 (3) poorer shielding of 5d electrons by 4f electrons
 (4) greater shielding of 5d electrons by 4f electrons
 13. Which one of the following sets of ions represents a collection of isoelectronic species ? **[AIIEEE-2006]**
 (1) N^{3-} , O^{2-} , F^- , S^{2-} (2) Li^+ , Na^+ , Mg^{+2} , Ca^{+2}
 (3) K^+ , Cl^- , Ca^{+2} , Sc^{+3} (4) Ba^{+2} , Sr^{+2} , K^{+2} , Ca^{+2}
 14. The increasing order of the first ionization enthalpies of the elements B, P, S and F (lowest first) is:- **[AIIEEE-2006]**
 (1) $\text{F} < \text{S} < \text{P} < \text{B}$ (2) $\text{P} < \text{S} < \text{B} < \text{F}$
 (3) $\text{B} < \text{P} < \text{S} < \text{F}$ (4) $\text{B} < \text{S} < \text{P} < \text{F}$
 15. Which one of the following constitutes a group of the isoelectronic species? **[AIIEEE-2008]**
 (1) C_2^{2-} , O_2^- , CO , NO (2) NO^+ , C_2^{2-} , CN^- , N_2
 (3) CN^- , N_2 , O_2^{2-} , C_2^{2-} (4) N_2 , O_2^- , NO^+ , CO
 16. The set representing the correct order of ionic radius is :- **[AIIEEE-2009]**
 (1) $\text{Li}^+ > \text{Na}^+ > \text{Mg}^{2+} > \text{Be}^{2+}$
 (2) $\text{Mg}^{2+} > \text{Be}^{2+} > \text{Li}^+ > \text{Na}^+$
 (3) $\text{Li}^+ > \text{Be}^{2+} > \text{Na}^+ > \text{Mg}^{2+}$
 (4) $\text{Na}^+ > \text{Li}^+ > \text{Mg}^{2+} > \text{Be}^{2+}$

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CHEMICAL BONDING

EXERCISE-I

- Which one of the following pairs of molecules will have permanent dipole moments for both members?
(1) NO_2 and CO_2 (2) NO_2 and O_3
(3) SiF_4 and CO_2 (4) SiF_4 and NO_2
- The pair of species having identical shapes for molecules of both species is :
(1) XeF_2 , CO_2 (2) BF_3 , PCl_3
(3) PF_5 , IF_5 (4) CF_4 , SF_4
- The correct order of bond angles (smallest first) in H_2S , NH_3 , BF_3 and SiH_4 is :-
(1) $\text{H}_2\text{S} < \text{NH}_3 < \text{SiH}_4 < \text{BF}_3$
(2) $\text{NH}_3 < \text{H}_2\text{S} < \text{SiH}_4 < \text{BF}_3$
(3) $\text{H}_2\text{S} < \text{SiH}_4 < \text{NH}_3 < \text{BF}_3$
(4) $\text{H}_2\text{S} < \text{NH}_3 < \text{BF}_3 < \text{SiH}_4$
- lattice energy of an ionic compound depends upon
(1) charge on the ion only
(2) size of the ion only
(3) packing of the ion only
(4) charge and size of the ion
- The correct order of increasing C - O bond length of, CO , CO_3^{2-} , CO_2 is
(1) $\text{CO}_3^{2-} < \text{CO}_2 < \text{CO}$
(2) $\text{CO}_2 < \text{CO}_3^{2-} < \text{CO}$
(3) $\text{CO} < \text{CO}_3^{2-} < \text{CO}_2$
(4) $\text{CO} < \text{CO}_2 < \text{CO}_3^{2-}$
- Amongst H_2O , H_2S , H_2Se and H_2Te , the one with the highest boiling point is
(1) H_2O because of hydrogen bonding
(2) H_2Te because of higher molecular weight
(3) H_2S because of hydrogen bonding
(4) H_2Se because of lower molecular weight
- Which species has the maximum number of lone pair of electrons on the central atom?
(1) ClO_3^- (2) XeF_4
(3) SF_4 (4) I_3^-
- Which is not an exception to octet rule ?
(1) BF_3 (2) SnCl_4
(3) BeI_2 (4) ClO_2
- The correct order of decreasing bond energy is:-
(1) $\text{O}-\text{O} > \text{S}-\text{S} > \text{Se}-\text{Se}$
(2) $\text{C}-\text{C} > \text{Si}-\text{Si} > \text{Ge}-\text{Ge}$
(3) $\text{F}-\text{F} > \text{O}-\text{O} > \text{N}-\text{N}$
(4) $\text{F}-\text{F} > \text{Cl}-\text{Cl} > \text{Br}-\text{Br}$
- Both CO_2 and H_2O contain polar covalent bonds but CO_2 is nonpolar while H_2O is polar because-
(1) H atom is smaller than C atom
(2) CO_2 is a linear molecule while H_2O is an angular molecule
(3) O - H bond is more polar than C - H bond
(4) CO_2 contains multiple bonds while H_2O has only single bonds
- Maximum bond energy will be shown by the species
(1) O_2^+ (2) O_2 (3) O_2^- (4) O_2^{2-}
- Resonance is not shown by -
(1) C_6H_6 (2) CO_2 (3) CO_3^{2-} (4) SiO_2
- Two ice cubes are pressed over each other until they unite to form one block. The force mainly responsible for holding them together is-
(1) van der Waals force
(2) dipole-dipole interaction
(3) H bonding
(4) covalent bonding
- Which order of decreasing bond angle is correct?
(1) $\text{CCl}_4 > \text{BF}_3 > \text{NO}_2^+$
(2) $\text{NH}_3 > \text{NCl}_3 > \text{NBr}_3$
(3) $\text{Br}_2\text{O} > \text{Cl}_2\text{O} > \text{OF}_2$
(4) $\text{PCl}_3 > \text{PBr}_3 > \text{PI}_3$
- On the basis of hybridization of one s & one p orbitals they are arrange at :-
(1) Two orbitals mutually at 90° angle
(2) two orbitals mutually at 180° angle
(3) Two orbitals mutually at 120° angle
(4) Two orbitals mutually at 150° angle
- The shape of IF_4^+ will be :-
(1) Square planar
(2) Tetranedral
(3) Pentagonal bipyramidal
(4) Distorted tetrahedral
- Which of the following has the highest value of dipole moment ?
(1) HCl (2) HF (3) HI (4) HBr
- The AsF_5 molecule is trigonal bipyramidal. The hybrid orbitals used by the As atoms for bonding are:
(1) $d_{x^2-y^2}, d_{z^2}, s, p_x, p_y$ (2) d_{xy}, s, p_x, p_y, p_z
(3) $s, p_x, p_y, p_z, d_{z^2}$ (4) $d_{x^2-y^2}, s, p_x, p_y$
- Which of the following is paramagnetic ?
(1) O_2^- (2) CN^- (3) CO (4) NO^+

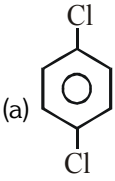
20. Acetic acid exists as dimer in benzene due to:-
 (1) Condensation reaction
 (2) Hydrogen bonding
 (3) Presence of carboxyl group
 (4) None of the above
21. Which of the following compound gives paramagnetic gas on heating ?
 (1) LiNO_3 (2) NaNO_3
 (3) KNO_3 (4) All of these
22. Which of the following statement is correct ?
 As the %s-character of a hybrid orbital decreases
 (1) The bond angle decreases
 (2) The bond strength increases
 (3) The bond length decreases
 (4) Size of orbital decreases
23. In S_8 each sulphur atom is :-
 (1) sp hybridised with a planar ring
 (2) sp^3 hybridised with a planar ring
 (3) sp^3 hybridised with a non-planar ring
 (4) sp^3d hybridised two sulphur atoms
24. Maximum number of identical bond length are present in
 (1) SF_6 (2) IF_7 (3) PCl_5 (4) SO_4^{2-}
25. Which of the following molecule is planar as well as polar ?
 (1) PCl_3 (2) SF_4
 (3) ClF_3 (4) None of these
26. Which one is formed in II^{nd} excited state ?
 (1) PCl_5 (2) SH_6
 (3) SO_3 (4) IF_7
27. In which of the following compounds, breaking of covalent bond take place :-
 (1) Boiling of H_2O (2) Melting of KCN
 (3) Boiling of CF_4 (4) Melting of SiO_2
28. Which of the following order is incorrect ?
 (1) Ionic character = $\text{MCl} < \text{MCl}_2 < \text{MCl}_3$
 (2) Polarisability = $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$
 (3) Polarising power = $\text{Na}^+ < \text{Ca}^{+2} < \text{Mg}^{+2} < \text{Al}^{+3}$
 (4) Covalent character = $\text{LiF} < \text{LiCl} < \text{LiBr} < \text{LiI}$
29. The salt having least solubility in water :-
 (1) BaCl_2 (2) $\text{Ba}(\text{NO}_3)_2$
 (3) MgSO_4 (4) BaSO_4
30. Which of the following substance on being heated will give a gas that does not turn lime water milky?
 (1) LiNO_3 (2) ZnCO_3
 (3) ZnSO_3 (4) MgCO_3

				ANSWER KEY			Exercise-I			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	1	4	4	1	4	2	2	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	3	3	2	4	2	3	1	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	1	3	1	3	3	4	1	4	1

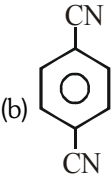
PREVIOUS YEARS' QUESTIONS

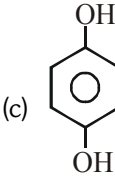
EXERCISE-II

- Which of the following molecules\ions does not contain unpaired electrons? [AIEEE-2006]
(1) N_2^+ (2) O_2 (3) O_2^{2-} (4) B_2
- Among the following mixtures, dipole-dipole as the major interaction, is present in [AIEEE-2006]
(1) KCl and water
(2) benzene and carbon tetrachloride
(3) benzene and ethanol
(4) acetonitrile and acetone
- In which of the following molecules/ions are all the bonds **not** equal? [AIEEE-2006]
(1) XeF_4 (2) BF_4^-
(3) SF_4 (4) SiF_4
- The decreasing values of bond angles from NH_3 (106°) to SbH_3 (91°) down group-15 of the periodic table is due to [AIEEE-2006]
(1) decreasing lp – bp repulsion
(2) increasing electronegativity
(3) increasing bp – bp repulsion
(4) increasing p-orbital character in sp^3
- In which of the following ionization processes, the bond order has increased and the magnetic behaviour has changed [AIEEE-2007]
(1) $NO \rightarrow NO^+$ (2) $O_2 \rightarrow O_2^+$
(3) $N_2 \rightarrow N_2^+$ (4) $C_2 \rightarrow C_2^+$
- Which of the following hydrogen bonds is the strongest [AIEEE-2007]
(1) $F-H \cdots F$ (2) $O-H \cdots O$
(3) $O-H \cdots F$ (4) $O-H \cdots N$
- Which of the following species exhibits the diamagnetic behaviour [AIEEE-2007]
(1) O_2^+ (2) O_2 (3) NO (4) O_2^{2-}
- The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represent the increasing order of the polarizing power of the cationic species, K^+ , Ca^{+2} , Mg^{+2} , Be^{+2} [AIEEE-2007]
(1) $Be^{+2} < K^+ < Ca^{+2} < Mg^{+2}$
(2) $K^+ < Ca^{+2} < Mg^{+2} < Be^{+2}$
(3) $Ca^{+2} < Mg^{+2} < Be^{+2} < K^+$
(4) $Mg^{+2} < Be^{+2} < K^+ < Ca^{+2}$
- Using MO theory predict which of the following species has the shortest bond length? [AIEEE-2009]
(1) O_2^- (2) O_2^{2-} (3) O_2^{+} (4) O_2^+
- The hybridisation of orbitals of N atom in NO_3^- , NO_2^+ and NH_4^+ are respectively:- [AIEEE-2011]
(1) sp , sp^3 , sp^2 (2) sp^2 , sp^3 , sp
(3) sp , sp^2 , sp^3 (4) sp^2 , sp , sp^3
- The structure of IF_7 is :- [AIEEE-2011]
(1) octahedral
(2) pentagonal bipyramid
(3) square pyramid
(4) trigonal bipyramid
- Among the following the maximum covalent character is shown by the compound :- [AIEEE-2011]
(1) $AlCl_3$ (2) $MgCl_2$
(3) $FeCl_2$ (4) $SnCl_2$
- Which of the following has maximum number of lone pairs associated with Xe ? [AIEEE-2011]
(1) XeO_3 (2) XeF_4
(3) XeF_6 (4) XeF_2
- The number of types of bonds between two carbon atoms in calcium carbide is :- [AIEEE-2005, 2011]
(1) One sigma, two pi (2) One sigma, one pi
(3) Two sigma, one pi (4) Two sigma, two pi
- Ortho-Nitrophenol is less soluble in water than p- and m- Nitrophenols because :- [AIEEE-2005, 2012]
(1) Melting point of o-Nitrophenol is lower than those of m- and p- isomers
(2) o-Nitrophenol is more volatile in steam than those of m- and p- isomers
(3) o-Nitrophenol shows Intramolecular H-bonding
(4) o-Nitrophenol shows Intermolecular H-bonding
- The molecule having smallest bond angle is :- [AIEEE-2012]
(1) PCl_3 (2) NCl_3
(3) $AsCl_3$ (4) $SbCl_3$
- In which of the following pairs the two species are not isostructural ? [AIEEE-2012]
(1) AlF_6^{3-} and SF_6 (2) CO_3^{2-} and NO_3^-
(3) PCl_4^+ and $SiCl_4$ (4) PF_5 and BrF_5

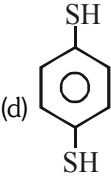
- 18.** Among the following species which two have trigonal bipyramidal shape ? [AIEEE-2012 (Online)]
 (I) NI_3 (II) I_3^-
 (III) SO_3^{2-} (IV) NO_3^-
 (1) II and III (2) III and IV
 (3) I and IV (4) None of them
- 19.** Among the following, the species having the smallest bond is :- [AIEEE-2012 (Online)]
 (1) NO (2) NO^+ (3) O_2 (4) NO^-
- 20.** Based on lattice energy and other considerations, which one of the following alkali metal chloride is expected to have the highest melting point ? [AIEEE-2012 (Online)]
 (1) RbCl (2) LiCl (3) KCl (4) NaCl
- 21.** Which of the following has the square planar structure ? [AIEEE-2012 (Online)]
 (1) NH_4^+ (2) CCl_4
 (3) XeF_4 (4) BF_4^-
- 22.** The compound of Xenon with zero dipole moment is :- [AIEEE-2012 (Online)]
 (1) XeO_3 (2) XeO_2
 (3) XeF_4 (4) XeOF_4
- 23.** Among the following the molecule with the lowest dipole moment is :- [AIEEE-2012 (Online)]
 (1) CHCl_3 (2) CH_2Cl_2
 (3) CCl_4 (4) CH_3Cl
- 24.** The formation of molecular complex $\text{BF}_3 - \text{NH}_3$ results in a change in hybridisation of boron :- [AIEEE-2012 (Online)]
 (1) from sp^3 to sp^3d
 (2) from sp^2 to dsp^2
 (3) from sp^3 to sp^2
 (4) from sp^2 to sp^3
- 25.** Which one of the following molecules is expected to exhibit diamagnetic behaviour ? [JEE (MAIN) 2013]
 (1) C_2 (2) N_2
 (3) O_2 (4) S_2
- 26.** In which of the following pairs of molecules/ions, both the species are not likely to exist ? [JEE (MAIN) 2013]
 (1) H_2^+ , He_2^{2-} (2) H_2^- , He_2^{2-}
 (3) H_2^{2+} , He_2 (4) H_2^- , He_2^{2+}
- 27.** Stability of the species Li_2 , Li_2^- and Li_2^+ increases in the order of :- [JEE (MAIN) 2013]
 (1) $\text{Li}_2 < \text{Li}_2^+ < \text{Li}_2^-$
 (2) $\text{Li}_2^- < \text{Li}_2^+ < \text{Li}_2$
 (3) $\text{Li}_2 < \text{Li}_2^- < \text{Li}_2^+$
 (4) $\text{Li}_2^- < \text{Li}_2 < \text{Li}_2^+$
- 28.** Which one of the following properties is **not** shown by NO ? [JEE (MAIN) 2014]
 (1) It combines with oxygen to form nitrogen dioxide
 (2) Its bond order is 2.5
 (3) It is diamagnetic in gaseous state
 (4) It is a neutral oxide
- 29.** For which of the following molecule significant $\mu \neq 0$? [JEE (MAIN) 2014]
- 

(a)



(b)
- 

(c)



(d)
- (1) Only (c) (2) (c) and (d)
 (3) Only (a) (4) (a) and (b)
- 30.** Which of the following species is not paramagnetic :- [JEE (MAIN) 2017]
 (1) NO (2) CO
 (3) O_2 (4) B_2

PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	4	3	4	1	1	4	2	3	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	4	1	3	4	4	4	2	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	3	4	1, 2	3	2	3	2	2

p-BLOCK ELEMENTS

EXERCISE-I

1. In which of the dimerisation process, the achievement of the octet is not the driving force.

- (1) $2\text{AlCl}_3 \longrightarrow \text{Al}_2\text{Cl}_6$
- (2) $\text{BeCl}_2 \longrightarrow \text{BeCl}_2 \text{ (solid)}$
- (3) $2\text{ICl}_3 \longrightarrow \text{I}_2\text{Cl}_6$
- (4) $2\text{NO}_2 \longrightarrow \text{N}_2\text{O}_4$

2. Column I

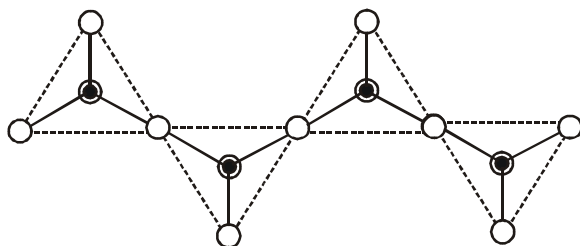
- (A) B_2H_6
- (B) Be_2H_4
- (C) Be_2Cl_4

Column II

- (P) $(3\text{C}-4\text{e}^-)$ bond
- (Q) $(3\text{C}-2\text{e}^-)$ bond
- (R) Vacant orbital participation in hybridisation
- (S) sp^3 hybridisation
- (T) sp^2 hybridisation

- (1) (A) - Q,R,S; (B) - Q,R,T; (C) - P,R,T; (D) - Q,R,S
- (2) (A) - Q,R,S,T; (B) - R,T; (C) - P,R,T; (D) - R,S
- (3) (A) - S,T; (B) - R,T; (C) - P,R,T; (D) - R,S,T
- (4) (A) - Q,S,T; (B) - R,T; (C) - P,R; (D) - R,S,T

3. A mineral contain following tetrameric anion in which $\bullet$ = Si, $\bigcirc$ = oxygen

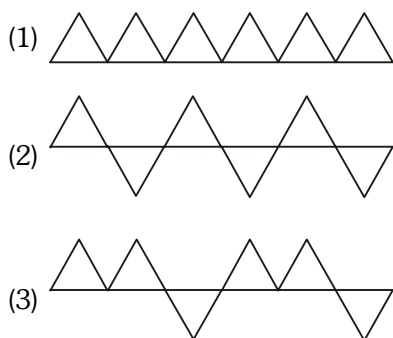


Select **correct** option (s) about anion in mineral-

- (1) Formula of anion is $(\text{SiO}_3)_n^{2n-}$ (where $n = 4$).
- (2) The total 10 negative charges are present in this anion.
- (3) It has three shared oxygen/corners and ten unshared oxygen/corners.
- (4) It is non planar

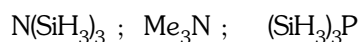
4. Silicate are existing mainly in the polymeric form. Several categories are available with us which depend on the mode of sharing of corners of SiO_4^{4-} tetrahedron.

Which of the following pyroxene chain silicate is having same formula.



- (4) All of these

5. The geometry with respect to the central atom of the following molecules are:



- (1) planar, pyramidal, planar
- (2) planar, pyramidal, pyramidal
- (3) pyramidal, pyramidal, pyramidal
- (4) pyramidal, planar, pyramidal

6. Column-I

- (A) $\text{N}(\text{SiH}_3)_3$
- (B) $\text{N}(\text{CH}_3)_3$
- (C) B_2H_6
- (D) BF_3

Column-II

- (P) $\text{p}\pi\text{-d}\pi$ back bonding
- (Q) sp^3 hybridisation for underlined atom
- (R) $\text{p}\pi\text{-p}\pi$ back bonding
- (S) neither $\text{p}\pi\text{-p}\pi$ nor $\text{p}\pi\text{-d}\pi$ back bonding
- (T) Underlined atom combine with electron rich molecule

- (1) (A) - P; (B) - Q,S; (C) - Q,S,T; (D) - R,T
- (2) (A) - P,Q; (B) - Q,S,T; (C) - R,T; (D) - S,T
- (3) (A) - P,Q; (B) - R,T; (C) - S,T; (D) - R,S,T
- (4) (A) - R,S,T; (B) - Q,R,S,T; (C) - S,T; (D) - P,Q,S,T

7. Choose the correct on the Cl-O bond length in NaClO_4 .

- (1) All Cl-O bonds are of equal length.
- (2) Three Cl-O bonds are of equal of length one longer.
- (3) Two Cl-O bonds are of same length which are longer compound to other two Cl-O bond length.
- (4) All Cl-O bond lengths are different

8. Column I

(Pair of species)

- (A) PCl_3F_2 , PCl_2F_3
- (B) BF_3 & BCl_3
- (C) CO_2 & CN_2^{-2}
- (D) C_6H_6 & $\text{B}_3\text{N}_3\text{H}_6$

Column II

(Identical Property in pairs of species)

- (P) Hybridisation of central atom
- (Q) Shape of molecule / ion
- (R) μ (dipole moment)
- (S) Total number of electrons

- (1) (A) - P,Q; (B) - P,Q,R; (C) - P,Q,R,S; (D) - P,Q,R,S
- (2) (A) - P,Q,R,S; (B) - P,Q; (C) - R,S; (D) - P,Q
- (3) (A) - P,Q; (B) - S,R; (C) - Q,R,S; (D) - R,S
- (4) (A) - P,Q; (B) - S,R; (C) - P,R; (D) - P,Q,R

9. Match the Column :

Column-I(A) $(\text{CH}_3)_2\text{PF}_3$ (B) SF_4 (C) XeF_4 (D) BrF_3 **Column-II**

(P) Lone pair of electrons on central atom = 1

(Q) Central atom is sp^3d hybridised

(R) Equal length of all central atom-fluorine bond

(S) Total number of lone pair of electrons is more than 9

(1) (A) – Q; (B) – P, Q, S; (C) – R, S; (D) – Q, S

(2) (A) – P, Q; (B) – P, Q, S; (C) – S, R; (D) – P, S

(3) (A) – R, S; (B) – P, Q, S; (C) – S, R; (D) – P, S

(4) (A) – Q, S; (B) – P, Q, S; (C) – S, R; (D) – P, Q, R, S

10. Select correct statement about hydrolysis of BCl_3 and NCl_3

(1) NCl_3 is hydrolysed and gives HOCl but BCl_3 is not hydrolysed.(2) Both NCl_3 and BCl_3 on hydrolysis gives HCl (3) NCl_3 on hydrolysis gives HOCl but BCl_3 gives HCl (4) Both NCl_3 and BCl_3 on hydrolysis gives HOCl

11. Which of the following statements are correct for SOF_4 molecule.

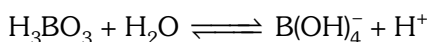
(1) It is square pyramidal in shape

(2) On hydrolysis it produces H_2SO_4 and HF

(3) All S–F bond lengths are of identical length

(4) Two S–F bond lengths are longer compared to other two S–F bond lengths

12. **Statement-1** : H_3BO_3 in water behaves as monobasic acid.

Statement-2 : The ionisation reaction is:

(1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(2) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(3) Statement-1 is true, statement-2 is false.

(4) Statement-1 is false, statement-2 is true.

13. For H_3PO_3 and H_3PO_4 , the correct choice is

(1) H_3PO_3 is dibasic and reducing agent.(2) H_3PO_3 is dibasic and non reducing agent.(3) H_3PO_4 is tribasic and reducing agent(4) H_3PO_4 is tribasic and non reducing agent.

14. Which of the following statement is **incorrect**?

(1) Oxidizing power order : $\text{SiCl}_4 < \text{SnCl}_4 < \text{PbCl}_4$ (2) Ionic character order : $\text{CsBr} > \text{RbBr} > \text{KBr} > \text{NaBr} > \text{LiBr}$

(3) The ionic character of lead (II) halides decreases with increase in atomic no. of halogen

(4) The oxidation state of Tl in TlI_3 is +3.

15. Choose the correct statement regarding bond angle:-

(1) $\widehat{\text{FCF}}$ in $\text{F}_2\text{CO} < \widehat{\text{HCH}}$ in H_2CO (2) $\widehat{\text{BrPBr}}$ in $\text{PBr}_3 < \widehat{\text{FPF}}$ in PF_3 (3) $\widehat{\text{FSF}}(\text{eq}) > \widehat{\text{FSF}}(\text{ax})$ in SF_4 (4) All $\widehat{\text{FIF}}$ angles in IF_5 are identical

16. **Column I**

(A) Dithionous acid

Column II

(P) S–O–S bond is not present

(B) Thiosulphuric acid

(Q) All S atom in the molecule has oxidation state +3

(C) Caro's acid

(R) Acidic strength of –OH groups present in the molecule is different

(D) Pyrosulphurous acid

(S) at least one S atom has oxidation state +5 in molecule

(1) (A) – P, Q; (B) – P; (C) – P, Q; (D) – P, R, S

(2) (A) – P; (B) – P, Q; (C) – P, R, S; (D) – P, Q, R, S

(3) (A) – P, Q; (B) – R, S; (C) – P, Q, S; (D) – P, Q, R, S

(4) (A) – P, Q, R, S; (B) – R, S; (C) – P, Q, R, S; (D) – P, Q

17. Structure of $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$ contains

(1) Two triangular and two tetrahedral units

(2) Three triangular and one tetrahedral units.

(3) All tetrahedral units.

(4) All triangular units.

18. Which of the following statement is incorrect :-

(1) The free electron of ClO_3 molecule is present in d-orbital of Cl-atom(2) The free electron of $\dot{\text{C}}\text{F}_3$ is present in sp^3 hybrid orbital

(3) NO is polar

(4) The free electron of ClO_2 molecule is present in d-orbital of Cl-atom

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

- Graphite is a soft solid lubricant extremely difficult to melt. The reason for this anomalous behaviour is that graphite : **[AIEEE-2003]**
 - Has molecules of variable molecular masses like polymers
 - Has carbon atoms arranged in large plates of rings of strongly bonded carbon atoms with weak interplate bonds
 - Is a non crystalline substance
 - Is an allotropic form of diamond
- The soldiers of Napoleon army while at Alps during freezing winter suffered a serious problem as regards to the tin buttons of their uniforms. White Metallic tin buttons get converted to grey powder. This transformation is related to:- **[AIEEE-2004]**
 - An interaction with water vapour contained in humid air
 - A change in crystalline structure of tin
 - A change in the partial pressure of O_2 in air
 - An interaction with N_2 of air at low temperature
- Which is the most thermodynamically stable allotropic form of phosphorus? **[IIT- 2004]**
 - Red
 - White
 - Black
 - Yellow
- The number of hydrogen atoms attached to phosphorus atom in hypophosphorous acid is : **[AIEEE-2005]**
 - Zero
 - Two
 - One
 - Three
- Which one of the following is the correct statement **[AIEEE-2005]**
 - Boric acid is a protonic acid
 - Beryllium exhibits coordination number of six
 - Chlorides of both beryllium and aluminium have bridged chloride structures in solid phase
 - B_2H_6 , $2NH_3$ is known as "inorganic benzene"
- In silicon dioxide : **[AIEEE-2005]**
 - Each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms
 - Each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms
 - Silicon atom is bonded to two oxygen atoms
 - There are double bonds between silicon and oxygen atoms
- Which of the following is not oxidised by O_3 ? **[IIT- 2005]**
 - KI
 - $FeSO_4$
 - $KMnO_4$
 - K_2MnO_4
- When PbO_2 reacts with conc. HNO_3 the gas evolved may be : **[IIT 2005]**
 - NO_2
 - O_2
 - N_2
 - N_2O
- The stability of dihalides of Si, Ge, Sn and Pb increases steadily in the sequence: **[AIEEE-2007]**
 - $GeX_2 << SiX_2 << SnX_2 << PbX_2$
 - $SiX_2 << GeX_2 << PbX_2 << SnX_2$
 - $SiX_2 << GeX_2 << SnX_2 << PbX_2$
 - $PbX_2 << SnX_2 << GeX_2 << SiX_2$
- Among the following, the paramagnetic compound is - **[IIT- 2007]**
 - Na_2O_2
 - O_3
 - N_2O
 - KO_2
- Among the following substituted silanes the one which will give rise to cross linked silicone polymer on hydrolysis is **[AIEEE-2008]**
 - R_4Si
 - $RSiCl_3$
 - R_2SiCl_2
 - R_3SiCl
- Which of the following statements regarding sulphur is incorrect ? **[AIEEE-2011]**
 - At $600^\circ C$ the gas mainly consists of S_2 molecules
 - The oxidation state of sulphur is never less than +4 in its compounds
 - S_2 molecule is paramagnetic
 - The vapour at $200^\circ C$ consists mostly of S_8 rings
- The number of S-S bonds in SO_3 , $S_2O_3^{2-}$, $S_2O_6^{2-}$ and $S_2O_8^{2-}$ respectively are :- **[JEE Main(Online)-2012]**
 - 1, 0, 1, 0
 - 0, 1, 1, 0
 - 1, 0, 0, 1
 - 0, 1, 0, 1
- Which one of the following depletes ozone layer ? **[JEE Main(Online)-2012]**
 - NO and freons
 - SO_2
 - CO
 - CO_2
- The formation of molecular complex $BF_3 - NH_3$ results in a change in hybridisation of boron :- **[JEE(Main) Online-2012]**
 - from sp^3 to sp^3d
 - from sp^2 to dsp^2
 - from sp^3 to sp^2
 - from sp^2 to sp^3
- Which of the following xenon-OXO compounds may not be obtained by hydrolysis of xenon fluorides ? **[JEE Main(Online)-2014]**
 - XeO_2F_2
 - XeO_3
 - XeO_4
 - $XeOF_4$
- Consider the reaction **[JEE Main(Online)-2014]**

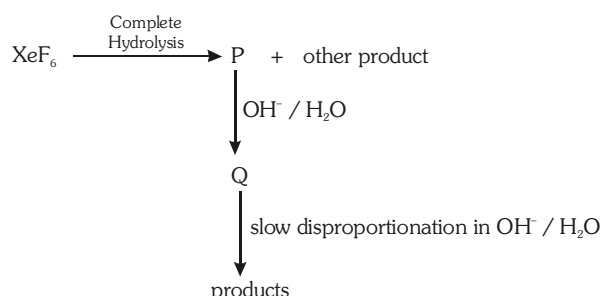
$$H_2SO_{3(aq)} + Sn_{(aq)}^{4+} + H_2O_{(l)} \rightarrow Sn_{(aq)}^{2+} + HSO_{4(aq)}^- + 3H_{(aq)}^+$$

Which of the following statements is correct?

 - H_2SO_3 is the reducing agent because it undergoes oxidation
 - H_2SO_3 is the reducing agent because it undergoes reduction
 - Sn^{4+} is the reducing agent because it undergoes oxidation
 - Sn^{4+} is the oxidizing agent because it undergoes oxidation

18. Under ambient conditions, the total number of gases released as products in the final step of the reaction scheme shown below is

[JEE Adv. 2014]



- (1) 0 (2) 1 (3) 2 (4) 3
19. Which of the following compounds has a P-P bond:-
[JEE Main(Online)-2015]
- (1) $\text{H}_4\text{P}_2\text{O}_5$ (2) $(\text{HPO}_3)_3$
(3) $\text{H}_4\text{P}_2\text{O}_7$ (4) $\text{H}_4\text{P}_2\text{O}_6$
20. Which among the following is the most reactive ?
[JEE Main-2015]
- (1) I_2 (2) ICl (3) Cl_2 (4) Br_2
21. From the following statements regarding H_2O_2 , choose the incorrect statement : [JEE Main-2015]
- (1) It has to be stored in plastic or wax lined glass bottles in dark
(2) It has to be kept away from dust
(3) It can act only as an oxidizing agent
(4) It decomposes on exposure to light
22. The reaction of zinc with dilute and concentrated nitric acid, respectively produces :
[JEE (Main) 2016]
- (1) NO_2 and N_2O (2) N_2O and NO_2
(3) NO_2 and NO (4) NO and N_2O
23. Which intermolecular force is most responsible in allowing xenon gas to liquefy?
[JEE (Main) Online 2016]
- (1) Ionic
(2) Instantaneous dipole- induced dipole
(3) Dipole - dipole
(4) Ion - dipole

24. The crystalline form of borax has
[JEE Adv. 2016]
- (1) Tetranuclear $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$ unit
(2) All boron atoms in the same plane
(3) Equal number of sp^2 and sp^3 hybridized boron atoms
(4) One terminal hydroxide per boron atom
25. Which of the following reactions is an example of a redox reaction ?
[JEE (Main) 2017]
- (1) $\text{XeF}_4 + \text{O}_2\text{F}_2 \rightarrow \text{XeF}_6 + \text{O}_2$
(2) $\text{XeF}_2 + \text{PF}_5 \rightarrow [\text{XeF}]^+\text{PF}_6^-$
(3) $\text{XeF}_6 + \text{H}_2\text{O} \rightarrow \text{XeOF}_4 + 2\text{HF}$
(4) $\text{XeF}_6 + 2\text{H}_2\text{O} \rightarrow \text{XeO}_2\text{F}_2 + 4\text{HF}$
26. The products obtained when chlorine gas reacts with cold and dilute aqueous NaOH are :-
[JEE (Main) 2017]
- (1) ClO^- and ClO_3^- (2) ClO_2^- and ClO_3^-
(3) Cl^- and ClO^- (4) Cl^- and ClO_2^-
27. The order of the oxidation state of the phosphorus atom in H_3PO_2 , H_3PO_4 , H_3PO_3 and $\text{H}_4\text{P}_2\text{O}_6$ is
[JEE Adv. 2017]
- (1) $\text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2$
(2) $\text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6$
(3) $\text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_4$
(4) $\text{H}_3\text{PO}_4 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6$
28. The option(s) with only amphoteric oxides is (are):
[JEE Adv. 2017]
- (1) Cr_2O_3 , CrO , SnO , PbO
(2) NO , B_2O_3 , PbO , SnO_2
(3) Cr_2O_3 , BeO , SnO , SnO_2
(4) ZnO , Al_2O_3 , PbO , PbO_2
29. The colour of the X_2 molecules of group 17 elements changes gradually from yellow to violet down the group. This is due to -
[JEE Adv. 2017]
- (1) the physical state of X_2 at room temperature changes from gas to solid down the group
(2) decrease in HOMO-LUMO gap down the group
(3) decrease in $\pi^*-\sigma^*$ down the group
(4) decrease in ionization energy down the group
30. Xenon hexafluoride on partial hydrolysis produces compounds 'X' and 'Y' Compounds 'X' and 'Y' and the oxidation state of Xe are respectively :
[JEE (Main) ONLINE 2018]
- (1) $\text{XeO}_2\text{F}_2(+6)$ and $\text{XeO}_2(+4)$
(2) $\text{XeOF}_4(+6)$ and $\text{XeO}_2\text{F}_2(+6)$
(3) $\text{XeOF}_4(+6)$ and $\text{XeO}_3(+6)$
(4) $\text{XeO}_2(+4)$ and $\text{XeO}_3(+6)$

PREVIOUS YEARS QUESTIONS				ANSWER KEY				Exercise-II		
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	3	2	3	1	3	2	3	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	2	1	4	3	1	3	4	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	2	2	1,3,4	1	3	1	3,4	2,3	2

d & f BLOCK ELEMENTS

EXERCISE-I

- Which element is having lowest melting and boiling point :
(1) Ti (2) Cu (3) Zn (4) Mn
- Which of the following set of metals can form alloy:
(1) Cu – Au (2) Li–Na
(3) Fe–Hg (4) All
- Not more than one oxidation state is show by :
(1) Mn (2) Cr (3) Fe (4) Sc
- The reaction of O₂ and CO with haemoglobin gives:
(1) Only oxygen-haeme complex
(2) Only CO-haeme complex
(3) Both but oxygen-haeme-complex is more stable
(4) Both but CO-haeme-complex is more stable
- Iron becomes passive by.....due to formation of..... :
(1) dil. HCl, Fe₂O₃ (2) Aqua-regia, Fe₃O₄
(3) conc. H₂SO₄, Fe₃O₄ (4) conc. HCl, Fe₃O₄

$$\text{MnO}_4^- + \text{xe}^- \longrightarrow \text{MnO}_4^{2-}$$

$$\begin{array}{l} \xrightarrow{+ \text{ye}^- \text{ (Acidic medium)}} \text{Mn}^{+2} \\ \xrightarrow{+ \text{ze}^- \text{ (Natural medium)}} \text{MnO}_2 \end{array}$$

x, y and z are respectively :
(1) 1, 2, 3 (2) 1, 5, 3 (3) 1, 3, 5 (4) 5, 3, 1
- In the equation : $\text{M} + 8 \text{CN}^- + 2 \text{H}_2\text{O} + \text{O}_2 \longrightarrow 4[\text{M}(\text{CN})_2]^- + 4 \text{OH}^-$, metal M is :
(1) Ag (2) Au (3) Cu (4) Hg
- Amongst CuF₂, CuCl₂ and CuBr₂ :
(1) Only CuF₂ is ionic
(2) Both CuCl₂ and CuBr₂ are covalent
(3) CuF₂ and CuCl₂ are ionic but CuBr₂ is covalent
(4) CuF₂, CuCl₂ as well as CuBr₂ are ionic
- A metal M which is not affected by strong acids like conc. HNO₃, conc. H₂SO₄ and conc. solution of alkalis like NaOH, KOH forms MCl₃ which finds use for toning in photography. The metal M is :
(1) Ag (2) Hg (3) Au (4) Cu
- When excess of sodium thiosulphate is added to dil. AgNO₃ solution a soluble compound X is formed. However, when dil. Na₂S₂O₃ solution is added to conc. AgNO₃ solution a white ppt. turning yellow and finally black ppt. of Y is obtained. Which is correct pair. :
(1) X is Ag₂S and Y is Na₃[Ag(S₂O₃)₂]
(2) X is Na₃[Ag(S₂O₃)₂] and Y is Ag₂S
(3) X is Ag₂S₂O₃ and Y is Ag₂S
(4) X is Ag₂S₂O₃ and Y is Na₃[Ag(S₂O₃)₂]
- Which of the following ion is not coloured ?
(1) Ni(DMG)₂ (2) [Co(SCN)₄]²⁻
(3) [Fe(H₂O)₅SCN]²⁺ (4) [Al(OH)₄]⁻
- The total spin and paramagnetism (B.M.) of ferrocyanide ion are respectively :
(1) 0, 2√6 (2) 5/2, √35
(3) 2, √24 (4) 1, 2√2
- Which is not correct about FeO_{0.94} ?
(1) It is non-stoichiometric compound
(2) Some of Fe²⁺ ions get replaced by as many two third Fe³⁺ ions
(3) It is metal excess solid
(4) It is metal deficient solid.
- The electronic configuration of Cu(II) is 3d⁹ whereas that of Cu(I) is 3d¹⁰. Which of the following is correct?
(1) Cu(II) is more stable
(2) Cu(II) is less stable
(3) Cu(I) and Cu(II) are equally stable
(4) Stability of Cu(I) and Cu(II) depends on nature of copper salts
- Generally transition elements form coloured salts due to the presence of unpaired electrons. Which of the following compounds will be coloured in solid state?
(1) Ag₂SO₄ (2) CuF₂ (3) ZnF₂ (4) Cu₂Cl₂
- Which of the following reactions are disproportionation reactions?
(a) $2\text{Cu}^+ \rightarrow \text{Cu}^{2+} + \text{Cu}$
(b) $3\text{MnO}_4^{2-} + 4\text{H}^+ \rightarrow 2\text{MnO}_4^- + \text{MnO}_2 + 2\text{H}_2\text{O}$
(c) $2\text{KMnO}_4 \rightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$
(d) $2\text{MnO}_4^- + 3\text{Mn}^{2+} + 2\text{H}_2\text{O} \rightarrow 5\text{MnO}_2 + 4\text{H}^+$
(1) a, b (2) a, b, c (3) b, c, d (4) a, d
- Which of the following oxidation state is common for all lanthanoids?
(1) +2 (2) +3 (3) +4 (4) +5
- There are 14 elements in actinoid series. Which of the following elements does not belong to this series?
(1) U (2) Np (3) Tm (4) Fm
- Which of the following is amphoteric oxide?
Mn₂O₇, CrO₃, Cr₂O₃, CrO, V₂O₅, V₂O₄
(1) V₂O₅, Cr₂O₃ (2) Mn₂O₇, CrO₃
(3) CrO, V₂O₅ (4) V₂O₅, V₂O₄
- Interstitial compounds are formed when small atoms are trapped inside the crystal lattice of metals. Which of the following is not the characteristic property of interstitial compounds?
(1) They have high melting points in comparison to pure metals.
(2) They are very hard.
(3) They retain metallic conductivity.
(4) They are chemically very reactive.
- When acidified K₂Cr₂O₇ solution is added to Sn²⁺ salts then Sn²⁺ changes to
(1) Sn (2) Sn³⁺ (3) Sn⁴⁺ (4) Sn⁺

- 22.** Highest oxidation state of manganese in fluoride is +4 (MnF_4) but highest oxidation state in oxides is +7 (Mn_2O_7) because _____.
 (1) fluorine is more electronegative than oxygen.
 (2) fluorine does not possess d-orbitals.
 (3) fluorine stabilises lower oxidation state.
 (4) in covalent compounds fluorine can form single bond only while oxygen forms double bond.
- 23.** Although Zirconium belongs to 4d transition series and Hafnium to 5d transition series even then they show similar physical and chemical properties because _____.
 (1) both belong to d-block.
 (2) both have same number of electrons.
 (3) both have similar atomic radius.
 (4) both belong to the same group of the periodic table.
- 24.** Why is HCl not used to make the medium acidic in oxidation reactions of KMnO_4 in acidic medium?
 (1) Both HCl and KMnO_4 act as oxidising agents.
 (2) KMnO_4 oxidises HCl into Cl_2 which is also an oxidising agent.
 (3) KMnO_4 is a weaker oxidising agent than HCl.
 (4) KMnO_4 acts as a reducing agent in the presence of HCl.
- 25.** In the form of dichromate, Cr(VI) is a strong oxidising agent in acidic medium but Mo(VI) in MoO_3 and W(VI) in WO_3 are not because _____.
 (1) Cr(VI) is more stable than Mo(VI) and W(VI).
 (2) Mo(VI) and W(VI) are more stable than Cr(VI).
 (3) Higher oxidation states of heavier members of group-6 of transition series are more stable.
 (4) Lower oxidation states of heavier members of group-6 of transition series are more stable.
- 26.** Which of the following actinoids show oxidation states upto +7?
 (1) Am (2) Pu (3) U (4) Np
- 27.** Which of the following will not act as oxidising agents?
 (1) CrO_3 (2) MoO_3 (3) WO_3 (4) CrO_4^{2-}
- 28.** Which of the following lanthanoids show +2 oxidation state besides the characteristic oxidation state +3 of lanthanoids?
 (1) Ce (2) Eu (3) Yb (4) Ho
- 29.** Match the property given in Column I with the element given in Column II.
- | Column I
(Property) | Column II
(Element) |
|---|------------------------|
| (A) Lanthanoid which shows +4 oxidation state | (i) Pm |
| (B) Lanthanoid which can show +2 oxidation state | (ii) Ce |
| (C) Radioactive lanthanoid | (iii) Lu |
| (D) Lanthanoid which has $4f^7$ electronic configuration in +3 oxidation state | (iv) Eu |
| (E) Lanthanoid which has $4f^{14}$ electronic configuration in +3 oxidation state | (v) Gd |
| | (vi) Dy |
- (1) (A) – ii; (B) – iv; (C) – i; (D) – v; (E) – iii
 (2) (A) – i; (B) – ii, iii; (C) – iv; (D) – i, v; (E) – v
 (3) (A) – iv; (B) – v; (C) – i, ii; (D) – i; (E) – iii
 (4) (A) – i; (B) – iv; (C) – i, ii; (D) – i; (E) – iii
- 30.** Match the properties given in Column I with the metals given in Column II.
- | Column I
(Property) | Column II
(Metal) |
|---|----------------------|
| (A) Element with highest second ionisation enthalpy | (i) Co |
| (B) Element with highest third ionisation enthalpy | (ii) Cr |
| (C) M in $\text{M}(\text{CO})_6$ is | (iii) Cu |
| (D) Element with highest heat of atomisation | (iv) Zn |
| | (v) Ni |
- (1) (A) – iii; (B) – iv; (C) – iii; (D) – i
 (2) (A) – i; (B) – iv; (C) – ii; (D) – iii
 (3) (A) – ii; (B) – i; (C) – iii; (D) – iv
 (4) (A) – iv; (B) – ii; (C) – i; (D) – iii

ANSWER KEY						Exercise-I				
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	4	4	2	2	2	1,2	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	3	1	2	1	2	3	1	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	4	3	2	2,3	2,4	2,3	2,3	1	1

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

- Which of the following alloys contains (s) Cu and Zn?
[JEE 1993]
(1) Bronze (2) Brass
(3) Gun metal (4) Type metal
- Which compound does not dissolve in hot, dilute HNO_3 ?
[JEE 1996]
(1) HgS (2) PbS
(3) CuS (4) CdS
- Ammonium dichromate is used in some fireworks. The green coloured powder blown in the air is -
[JEE 1997]
(1) CrO_3 (2) Cr_2O_3
(3) Cr (4) $\text{CrO}(\text{O}_2)$
- In the dichromate anion,
[JEE 1999]
(1) 4 Cr - O bonds are equivalent
(2) 6 Cr - O bonds are equivalent
(3) all Cr - O bonds are equivalent
(4) all Cr - O bonds are non-equivalent
- Anhydrous ferric chloride is prepared by:
[JEE 2002]
(1) heating hydrated ferric chloride at a high temperature in a stream of air
(2) heating metallic iron in a stream of dry chlorine gas
(3) reaction of ferric oxide with HCl
(4) reaction of metallic iron with HCl
- What would happen when a solution of potassium chromate is treated with an excess of dilute nitric acid -
[AIEEE-2003]
(1) Cr^{3+} and $\text{Cr}_2\text{O}_7^{2-}$ are formed
(2) $\text{Cr}_2\text{O}_7^{2-}$ and H_2O are formed
(3) $\text{Cr}_2\text{O}_7^{2-}$ is reduced to +3 state of Cr
(4) $\text{Cr}_2\text{O}_7^{2-}$ is oxidised to +7 state of Cr
- When MnO_2 is fused with KOH , a coloured compound is formed, the product and its colour is:
[JEE 2003]
(1) K_2MnO_4 , green
(2) KMnO_4 , purple
(3) Mn_2O_3 , brown
(4) Mn_3O_4 , black
- Calomel on reaction with NH_4OH gives
[AIEEE-2004]
(1) HgNH_2Cl (2) $\text{NH}_2\text{-Hg-Hg-Cl}$
(3) Hg_2O (4) HgO
- The product of oxidation of I^- with MnO_4^- in alkaline medium is -
[JEE 2004]
(1) IO_3^- (2) I_2
(3) IO^- (4) IO_4^-
- $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ on heating liberates a gas. The same gas will be obtained by -
[JEE 2004]
(1) heating NH_4NO_2
(2) heating NH_4NO_3
(3) treating H_2O_2 with NaNO_2
(4) treating Mg_3N_2 with H_2O
- Iron exhibits +2 and +3 oxidation states. Which of the following statements about iron is **incorrect**?
[AIEEE-2012]
(1) Ferrous compounds are more easily hydrolysed than the corresponding ferric compounds.
(2) Ferrous oxide is more basic in nature than the ferric oxide.
(3) Ferrous compounds are relatively more ionic than the corresponding ferric compounds.
(4) Ferrous compounds are less volatile than the corresponding ferric compounds.
- Which of the following arrangements does not represent the correct order of the property stated against it?
[JEE MAIN-2013]
(1) $\text{V}^{2+} < \text{Cr}^{2+} < \text{Mn}^{2+} < \text{Fe}^{2+}$: paramagnetic behaviour
(2) $\text{Ni}^{2+} < \text{Co}^{2+} < \text{Fe}^{2+} < \text{Mn}^{2+}$: ionic size
(3) $\text{Co}^{3+} < \text{Fe}^{3+} < \text{Cr}^{3+} < \text{Sc}^{3+}$: stability in aqueous solution
(4) $\text{Sc} < \text{Ti} < \text{Cr} < \text{Mn}$: number of oxidation states
- Consider the following reaction : [JEE MAIN-2013]
$$x\text{MnO}_4^- + y\text{C}_2\text{O}_4^{2-} + z\text{H}^+ \rightarrow x\text{Mn}^{2+} + 2y\text{CO}_2 + \frac{z}{2}\text{H}_2\text{O}$$

The values of x, y and z in the reaction are respectively :-
(1) 5, 2 and 16
(2) 2, 5 and 8
(3) 2, 5 and 16
(4) 5, 2 and 8

14. Potassium dichromate when heated with concentrated sulphuric acid and a soluble chloride, gives brown - red vapours of:

[JEE MAIN-2013, Online]

- (1) CrO_3 (2) Cr_2O_3
(3) CrCl_3 (4) CrO_2Cl_2

15. The element with which of the following outer electron configuration may exhibit the largest number of oxidation states in its compounds :

[JEE MAIN-2013, Online]

- (1) $3d^7 4s^2$ (2) $3d^8 4s^2$
(3) $3d^5 4s^2$ (4) $3d^6 4s^2$

16. When a small amount of KMnO_4 is added to concentrated H_2SO_4 , a green oily compound is obtained which is highly explosive in nature. Compound may be :

[JEE MAIN-2013, Online]

- (1) Mn_2O_3 (2) MnSO_4
(3) Mn_2O_7 (4) MnO_2

17. The equation which is balanced and represents the correct product (s) is :

[JEE MAIN-2014]

- (1) $[\text{Mg}(\text{H}_2\text{O})_6]^{2+} + (\text{EDTA})^{4-} \xrightarrow{\text{excess NaOH}} [\text{Mg}(\text{EDTA})]^{2-} + 6\text{H}_2\text{O}$
(2) $\text{CuSO}_4 + 4\text{KCN} \rightarrow \text{K}_2[\text{Cu}(\text{CN})_4] + \text{K}_2\text{SO}_4$
(3) $\text{Li}_2\text{O} + 2\text{KCl} \rightarrow 2\text{LiCl} + \text{K}_2\text{O}$
(4) $[\text{CoCl}(\text{NH}_3)_5]^+ + 5\text{H}^+ \rightarrow \text{Co}^{2+} + 5\text{NH}_4^+ + \text{Cl}^-$

18. Which series of reactions correctly represents chemical relations related to iron and its compound?

[JEE MAIN-2014]

- (1) $\text{Fe} \xrightarrow{\text{Cl}_2, \text{heat}} \text{FeCl}_3 \xrightarrow{\text{heat, air}} \text{FeCl}_2 \xrightarrow{\text{Zn}} \text{Fe}$
(2) $\text{Fe} \xrightarrow{\text{O}_2, \text{heat}} \text{Fe}_3\text{O}_4 \xrightarrow{\text{CO}, 600^\circ\text{C}} \text{FeO} \xrightarrow{\text{CO}, 700^\circ\text{C}} \text{Fe}$
(3) $\text{Fe} \xrightarrow{\text{dil H}_2\text{SO}_4} \text{FeSO}_4 \xrightarrow{\text{H}_2\text{SO}_4, \text{O}_2} \text{Fe}_2(\text{SO}_4)_3 \xrightarrow{\text{Heat}} \text{Fe}$
(4) $\text{Fe} \xrightarrow{\text{O}_2, \text{heat}} \text{FeO} \xrightarrow{\text{dil H}_2\text{SO}_4} \text{FeSO}_4 \xrightarrow{\text{Heat}} \text{Fe}$

19. Which of the following is **not** formed when H_2S reacts with acidic $\text{K}_2\text{Cr}_2\text{O}_7$ solution ?

[JEE MAIN-2014, Online]

- (1) K_2SO_4 (2) $\text{Cr}_2(\text{SO}_4)_3$
(3) S (4) CrSO_4

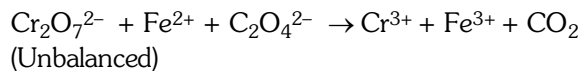
20. Which one of the following exhibits the largest number of oxidation states ?

[JEE MAIN-2014, Online]

- (1) Mn(25) (2) V(23) (3) Cr (24) (4) Ti (22)

21. How many electrons are involved in the following redox reaction ?

[JEE MAINS-2014, Online]



- (1) 3 (2) 4 (3) 5 (4) 6

22. Amongst the following, identify the species with an atom in +6 oxidation state:

[JEE MAIN-2014, Online]

- (1) $[\text{MnO}_4]^-$ (2) $[\text{Cr}(\text{CN})_6]^{3-}$
(3) Cr_2O_3 (4) CrO_2Cl_2

23. Copper becomes green when exposed to moist air for a long period. This is due to :-

[JEE MAIN-2014, Online]

- (1) the formation of a layer of cupric oxide on the surface of copper.
(2) the formation of basic copper sulphate layer on the surface of the metal
(3) the formation of a layer of cupric hydroxide on the surface of copper.
(4) the formation of a layer of basic carbonate of copper on the surface of copper.

24. Match the catalysts to the correct processes :-

[JEE MAIN-2015]

	Catalyst		Process
(A)	TiCl_3	(i)	Wacker process
(B)	PdCl_2	(ii)	Ziegler-Natta polymerization
(C)	CuCl_2	(iii)	Contact process
(D)	V_2O_5	(iv)	Deacon's process

- (1) A-ii, B-iii, C-iv, D-i
(2) A-iii, B-i, C-ii, D-iv
(3) A-iii, B-ii, C-iv, D-i
(4) A-ii, B-i, C-iv, D-iii

25. Which of the following statements is false :-

[JEE MAIN-2015, Online]

- (1) $\text{Cr}_2\text{O}_7^{2-}$ has a Cr - O - Cr bond
(2) CrO_4^{2-} is tetrahedral in shape
(3) $\text{Na}_2\text{Cr}_2\text{O}_7$ is a primary standard in volumetry
(4) $\text{K}_2\text{Cr}_2\text{O}_7$ is less soluble than $\text{Na}_2\text{Cr}_2\text{O}_7$

26. The correct statement(s) about Cr^{2+} and Mn^{3+} is (are)
[Atomic numbers of Cr = 24 and Mn = 25]

[JEE Advance 2015]

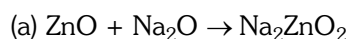
- (1) Cr^{2+} is a reducing agent
- (2) Mn^{3+} is an oxidizing agent
- (3) Both Cr^{2+} and Mn^{3+} exhibit d^4 electronic configuration
- (4) When Cr^{2+} is used as a reducing agent, the chromium ion attains d^5 electronic configuration

27. What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M ZnSO_4 ?

[JEE Main 2016]

- (1) The copper metal will dissolve and zinc metal will be deposited
- (2) No reaction will occur
- (3) The copper metal will dissolve with evolution of oxygen gas
- (4) The copper metal will dissolve with evolution of hydrogen gas

28. In the following reactions, ZnO is respectively acting as a/an : [JEE MAIN-2017]



- (1) base and acid
- (2) base and base
- (3) acid and acid
- (4) acid and base

29. Which of the following ions does **not** liberate hydrogen gas on reaction with dilute acids?

[JEE MAIN-2017, Online]

- | | |
|----------------------|----------------------|
| (1) Ti^{2+} | (2) Cr^{2+} |
| (3) Mn^{2+} | (4) V^{2+} |

30. Which of the following combination will produce H_2 gas ? [JEE Advance 2017]

- (1) Zn metal and NaOH(aq)
- (2) Au metal and NaCN(aq) in the presence of air
- (3) Cu metal and conc. HNO_3
- (4) Fe metal and conc. HNO_3

PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2,3	1	2	2	2	2	1	1	1	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	3	4	3	3	4	2	4	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	4	4	4	3	1,2,3	2	4	3	1

CO-ORDINATION CHEMISTRY

EXERCISE-I

- In the complex ion $[\text{Fe}(\text{EDTA})]^-$ the coordination number and oxidation state of central metal ion is :-
 (1) C. N. = 6 O. N. = +3
 (2) C. N. = 1 O. N. = -1
 (3) C. N. = 4 O. N. = +2
 (4) C. N. = 3 O. N. = +3
- Select bidentate or didentate ligand from the following .
 (1) CO (2) SCN^-
 (3) CH_3COO^- (4) $\text{C}_2\text{O}_4^{2-}$
- The oxidation and coordination number of Pt in $[\text{Pt}(\text{C}_2\text{H}_4)\text{Cl}_3]^-$ is respectively :-
 (1) + 1, 3 (2) + 2, 4
 (3) + 3, 6 (4) + 2, 5
- Give the IUPAC name of the complex compound $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}(\text{NO}_3)_2]$.
 (1) Bromoaquatetraamine Cobalt (III) nitrate
 (2) Bromoaquatetraaminocobalt (III) nitrate
 (3) Bromoaquatetraammine cobalt (III) nitrate
 (4) Tetraammineaquabromido cobalt (III) nitrate
- Which of the following complex is anion :-
 (1) Fluoro pentaammine cobalt (III)
 (2) Trioxalato ferrate (III)
 (3) Penta Carbonyl iron (0)
 (4) Dichloro diammine platinum
- The IUPAC name for $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ is :-
 (1) Hexaammine cobalt (III) hexacyanochromate (III)
 (2) Hexacyanochromium cobalt hexaammine (VI)
 (3) Hexaammine cobalt (III) hexacyanochromium (VI)
 (4) Hexacyanochromium (III) hexaammine cobalt (III)
- Which of the following has least conductivity in aqueous solution.
 (1) $\text{Co}(\text{NH}_3)_4\text{Cl}_3$ (2) $\text{Co}(\text{NH}_3)_3\text{Cl}_3$
 (3) $\text{Co}(\text{NH}_3)_5\text{Cl}_3$ (4) $\text{Co}(\text{NH}_3)_6\text{Cl}_3$
- The EAN of cobalt in the complex ion $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ is :-
 (1) 27 (2) 36 (3) 33 (4) 35
- Which gives only 25% mole of AgCl, when reacts with AgNO_3 :-
 (1) $\text{PtCl}_2 \cdot 4\text{NH}_3$ (2) $\text{PtCl}_4 \cdot 5\text{NH}_3$
 (3) $\text{PtCl}_4 \cdot 4\text{NH}_3$ (4) $\text{PtCl}_4 \cdot 3\text{NH}_3$
- Which of the following compound is paramagnetic
 (1) Tetracyanonickelate (II) ion
 (2) Tetraamminezinc (II) ion
 (3) Hexaamine chromium (III) ion
 (4) Diammine silver (I) ion
- The shape of the complex $[\text{Ag}(\text{NH}_3)_2]^+$ is :
 (1) Octahedral (2) Square planar
 (3) Tetrahedral (4) Linear
- Hexafluoroferrate (III) ion is an outer orbital complex. The number of unpaired electrons are
 (1) 1 (2) 5
 (3) 4 (4) Unpredictable
- The shape of $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$ is :
 (1) Square planar (2) Pyramidal
 (3) Octahedral (4) Tetrahedral
- The magnetic property and the shape of $[\text{Cr}(\text{NH}_3)_6]^{3+}$ complex ions are :
 (1) Paramagnetic, Octahedral
 (2) Diamagnetic, square planar
 (3) Paramagnetic, tetrahedral
 (4) None of the above
- In the complex $[\text{Ni}(\text{H}_2\text{O})_2(\text{NH}_3)_4]^{+2}$ the magnetic moment (μ) of Ni is :-
 (1) Zero (2) 2.83 BM
 (3) 1.73 BM (4) 3.87 BM
- Which of the following system has maximum number of unpaired electrons :-
 (1) d^5 (Octahedral, low spin)
 (2) d^8 (Tetrahedral)
 (3) d^6 (Octahedral, low spin)
 (4) d^3 (Octahedral)
- Match List-I (Complex ions) with List-II (Number of Unpaired Electrons) and select the correct answer using the codes given below the lists :-

List -I (Complex ions)	List II (Number of Unpaired Electrons)
A. $[\text{CrF}_6]^{4-}$	i. One
B. $[\text{MnF}_6]^{4-}$	ii. Two
C. $[\text{Cr}(\text{CN})_6]^{4-}$	iii. Three
D. $[\text{Mn}(\text{CN})_6]^{4-}$	iv. Four
	v. Five

Code :	A	B	C	D
(1)	iv	i	ii	v
(2)	ii	v	iii	i
(3)	iv	v	ii	i
(4)	ii	i	iii	v

18. Select most stable complex :-

- (1) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (2) $[\text{Co}(\text{NH}_3)_2(\text{en})_2]^{+3}$
 (3) $[\text{Co}(\text{en})_3]^{+3}$ (4) $[\text{Co}(\text{NH}_3)_4(\text{en})]^{+3}$

19. Which one of the following compounds will exhibit linkage isomerism :-

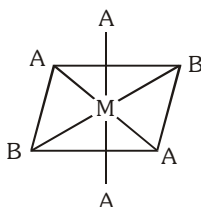
- (1) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (2) $[\text{Co}(\text{NH}_3)_2\text{NO}_2]\text{Cl}_2$
 (3) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (4) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$

20. Out of the following which complex will show geometrical isomerism ?

- (1) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (2) $\text{Ni}(\text{CO})_4$
 (3) $\text{Na}_3[\text{Ni}(\text{CN})_4]$ (4) $\text{K}[\text{Ag}(\text{CN})_2]$

21. Which of the following complexes will show optical isomerism ?

- (1) $[\text{Cr}(\text{NH}_3)_6]^{2+}$ (2) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{Pt}(\text{NH}_3)_3\text{Br}]\text{NO}_3$ (4) $[\text{Cr}(\text{en})_3]\text{Cl}_3$



22. The isomer -

can be marked as -

- (1) Cis isomer (2) Leavo isomer
 (3) Dextro isomer (4) Trans isomer

23. Which of the following complex does no show geometrical isomerism ?

- (1) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ (2) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$
 (3) $[\text{Cr}(\text{en})_3]^{3+}$ (4) $[\text{Pt}(\text{gly})_2]$

24. Which of the following is π -acid ligand

- (1) NH_3 (2) CO
 (3) gly. (4) ethylene diamine

25. Compounds which contain one or more metal carbon bonds are called :

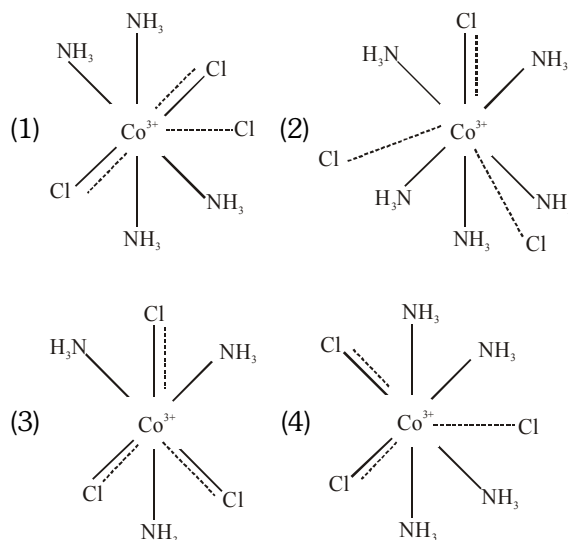
- (1) Organic compound
 (2) Complex compound
 (3) Metal carbides
 (4) OMC compounds.

26. A complex of platinum, ammonia and chloride produces four ions per molecule in the solution.

The structure consistent with the observation is:

- (1) $[\text{Pt}(\text{NH}_3)_4]\text{Cl}_4$ (2) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$
 (3) $[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$ (4) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$

27. Which of the following Werner's complex has least electrical conductivity ?



28. Which of the following complex shows ionization isomerism

- (1) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ (2) $[\text{Cr}(\text{en})_2]\text{Cl}_2$
 (3) $[\text{Cr}(\text{en})_3]\text{Cl}_3$ (4) $[\text{CoBr}(\text{NH}_3)_5]\text{SO}_4$

29. The number of unpaired electrons calculated in $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{CoF}_6]^{3-}$ are :

- (1) 4 and 4 (2) 0 and 2
 (3) 2 and 4 (4) 0 and 4

30. If $\lambda_{\text{absorbed}}$ for d-d transition is in order $[\text{Ti}(\text{X})_6]^{3+} > [\text{Ti}(\text{Y})_6]^{3+} > [\text{Ti}(\text{Z})_6]^{3+}$.

Select correct order of strength of ligands (X, Y, Z are monodentate ligand)-

- (1) $\text{Z} > \text{Y} > \text{X}$
 (2) $\text{X} > \text{Y} > \text{Z}$
 (3) $\text{Z} > \text{X} > \text{Y}$
 (4) Not predictable

ANSWER KEY						Exercise-I				
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	4	2	4	2	1	2	2	4	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	1	1	2	4	3	3	2	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	4	3	2	4	3	3	4	4	1

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

- The correct order of hybridisation of the central atom in the following species. [JEE 2001]
 NH_3 , $[\text{PtCl}_4]^{2-}$, PCl_5 and BCl_3 is [At No. Pt = 78]
 (1) $\text{dsp}^2, \text{sp}^3\text{d}, \text{sp}^2$ and sp^3
 (2) $\text{sp}^3, \text{dsp}^2$, sp^3d , sp^2
 (3) $\text{dsp}^2, \text{sp}^2, \text{sp}^3$ and sp^3d
 (4) dsp^2 , sp^3, sp^2 and sp^3d
- In $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$, the isomerism shown is - [AIEEE-2002]
 (1) Ligand (2) Optical
 (3) Geometrical (4) Ionization
- One mole of the complex compound $\text{Co}(\text{NH}_3)_5\text{Cl}_3$, gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of AgNO_3 solution to yield two moles of $\text{AgCl}(\text{s})$. The structure of the complex is - [AIEEE-2003]
 (1) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3] \cdot 2\text{NH}_3$ (2) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot \text{NH}_3$
 (3) $[\text{Co}(\text{NH}_3)_4\text{Cl}]\text{Cl}_2 \cdot \text{NH}_3$ (4) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
- In the coordination compound $\text{K}_4[\text{Ni}(\text{CN})_4]$, the oxidation state of nickel is - [AIEEE-2003]
 (1) 0 (2) +1 (3) +2 (4) -1
- The number of 3d-electrons remained in Fe^{2+} (At.no. of Fe = 26) ion is - [AIEEE-2003]
 (1) 4 (2) 5 (3) 6 (4) 3
- The coordination number of a central metal atom in a complex is determined by :- [AIEEE-2004]
 (1) The number of ligands around a metal ion bonded by sigma and pi-bonds both
 (2) The number of ligands around a metal ion bonded by pi-bonds
 (3) The number of ligands around a metal ion bonded by sigma bonds
 (4) The number of only anionic ligands bonded to the metal ion
- Which one of the following complexes is an outer orbital complex :- [AIEEE-2004]
 (1) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (2) $[\text{Mn}(\text{CN})_6]^{4-}$
 (3) $[\text{Fe}(\text{CN})_6]^{4-}$ (4) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
 (Atomic nos.: Mn=25 ; Fe=26 ; Co=27 ; Ni = 28)
- Coordination compounds have great importance in biological systems. In this context which of the following statements is incorrect ? [AIEEE-2004]
 (1) Cyanocobalamin is vitamin B_{12} and contains cobalt
 (2) Haemoglobin is the red pigment of blood and contains iron
 (3) Chlorophylls are green pigments in plants and contain calcium
 (4) Carboxypeptidase - A is an enzyme and contains zinc
- The correct order of magnetic moments (spin only values in B.M.) among is :- [AIEEE-2004]
 (1) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$
 (2) $[\text{MnCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$
 (3) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$
 (4) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
 (Atomic nos. : Mn = 25, Fe = 26, Co = 27)
- The species having tetrahedral shape is : [JEE 2004]
 (1) $[\text{PdCl}_4]^{2-}$ (2) $[\text{Ni}(\text{CN})_4]^{2-}$
 (3) $[\text{Pd}(\text{CN})_4]^{2-}$ (4) $[\text{NiCl}_4]^{2-}$
- The pair of compounds having metals in their highest oxidation state is [JEE 2004]
 (1) $\text{MnO}_2, \text{FeCl}_3$
 (2) $[\text{MnO}_4]^-$, CrO_2Cl_2
 (3) $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{CN})_3]$
 (4) $[\text{NiCl}_4]^{2-}$, $[\text{CoCl}_4]^-$
- For octahedral complex, the value of the 'spin only' magnetic moment for one of the following configurations is 2.84 BM. The correct one is [AIEEE-2005]
 (1) d^4 (in strong ligand field)
 (2) d^4 (in weak ligand field)
 (3) d^3 (in weak as well as in strong field)
 (4) d^5 (in strong ligand field)
- The IUPAC name for the complex $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ is - [AIEEE-2006]
 (1) pentaammine nitrito-N- cobalt (II) chloride
 (2) pentaammine nitrito-N- cobalt (III) chloride
 (3) nitrito-N- pentaamminecobalt (III) chloride
 (4) nitrito-N- pentaamminecobalt (II) chloride
- Nickel ($Z=28$) combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[\text{NiX}_4]^{2-}$. The number of unpaired electron in the nickel and geometry of this complex ion are, respectively. [AIEEE-2006]
 (1) one, square planar (2) two, square planar
 (3) one, tetrahedral (4) two, tetrahedral
- In $\text{Fe}(\text{CO})_5$, the Fe-C bond possesses [AIEEE-2006]
 (1) ionic character
 (2) σ -character only
 (3) π -character only
 (4) both σ and π character
- How many EDTA (ethylenediaminetetraacetate) molecules are required to make an octahedral complex with a Ca^{2+} ion ? [AIEEE-2006]
 (1) One (2) Two (3) Six (4) Three

17. The "spin-only" magnetic moment [in units of Bohr magneton, (μ_B)] of Ni^{2+} in aqueous solution would be (At. No. Ni = 28) [AIEEE-2006]
 (1) 0 (2) 1.73 (3) 2.84 (4) 4.90
18. Which one of the following has a square planar geometry :-
 (Co = 27, Ni = 28, Fe = 26, Pt = 78) [AIEEE-2007]
 (1) $[\text{CoCl}_4]^{2-}$ (2) $[\text{FeCl}_4]^{2-}$
 (3) $[\text{NiCl}_4]^{2-}$ (4) $[\text{PtCl}_4]^{2-}$
19. The coordination number and the oxidation state of the element 'E' in the complex $[\text{E}(\text{en})_2(\text{C}_2\text{O}_4^{2-})]\text{NO}_2^0$ (where en is ethylene diamine) are, respectively - [AIEEE-2008]
 (1) 6 and 2 (2) 4 and 2
 (3) 4 and 3 (4) 6 and 3
20. In which of the following octahedral complexes of Co (at. no. 27), will the magnitude of Δ_0 be the highest? [AIEEE-2008]
 (1) $[\text{Co}(\text{CN})_6]^{3-}$ (2) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$
 (3) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (4) $[\text{Co}(\text{NH}_3)_6]^{3+}$
21. Which of the following pairs represents linkage isomers? [AIEEE-2009]
 (1) $[\text{Co}(\text{NH}_3)_5\text{NO}_3]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{NO}_3$
 (2) $[\text{PtCl}_2(\text{NH}_3)_4]\text{Br}_2$ and $[\text{PtBr}_2(\text{NH}_3)_4]\text{Cl}_2$
 (3) $[\text{Cu}(\text{NH}_3)_4][\text{PtCl}_4]$ and $[\text{Pt}(\text{NH}_3)_4][\text{CuCl}_4]$
 (4) $[\text{Pd}(\text{PPh}_3)_2(\text{NCS})_2]$ and $[\text{Pd}(\text{PPh}_3)_2(\text{SCN})_2]$
22. Which of the following has an optical isomer? [AIEEE-2009]
 (1) $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$ (2) $[\text{Co}(\text{en})_2(\text{NH}_3)_2]^{3+}$
 (3) $[\text{Co}(\text{NH}_3)_3\text{Cl}]^+$ (4) $[\text{Co}(\text{en})(\text{NH}_3)_2]^{2+}$
23. Which one of the following has an optical isomer? [AIEEE-2010]
 (1) $[\text{Zn}(\text{en})_2]^{2+}$ (2) $[\text{Zn}(\text{en})(\text{NH}_3)_2]^{2+}$
 (3) $[\text{Co}(\text{en})_3]^{3+}$ (4) $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$
 (en = ethylenediamine)
24. A solution containing 2.675 g of $\text{CoCl}_3 \cdot 6\text{NH}_3$ (molar mass = 267.5 g mol⁻¹) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of AgNO_3 to give 4.78 g of AgCl (molar mass = 143.5 g mol⁻¹). The formula of the complex is :- [AIEEE-2010]
 (At. mass of Ag = 108 u)
 (1) $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$ (2) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
 (3) $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$ (4) $[\text{CoCl}_3(\text{NH}_3)_3]$
25. Which of the following facts about the complex $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ is wrong? [AIEEE-2011]
 (1) The complex is an outer orbital complex
 (2) The complex gives white precipitate with silver nitrate solution
 (3) The complex involves d^2sp^3 hybridisation and is octahedral in shape
 (4) The complex is paramagnetic
26. The magnetic moment (spin only) of $[\text{NiCl}_4]^{2-}$ is :- [AIEEE-2011]
 (1) 2.82 BM (2) 1.41 BM
 (3) 1.82 BM (4) 5.46 BM
27. Among the ligands NH_3 , en, CN^- and CO the correct order of their increasing field strength, is :- [AIEEE-2011]
 (1) $\text{CO} < \text{NH}_3 < \text{en} < \text{CN}^-$
 (2) $\text{NH}_3 < \text{en} < \text{CN}^- < \text{CO}$
 (3) $\text{CN}^- < \text{NH}_3 < \text{CO} < \text{en}$
 (4) $\text{en} < \text{CN}^- < \text{NH}_3 < \text{CO}$
28. Which one of the following complex ions has geometrical isomers? [AIEEE-2011]
 (1) $[\text{Co}(\text{en})_3]^{3+}$ (2) $[\text{Ni}(\text{NH}_3)_5\text{Br}]^+$
 (3) $[\text{Co}(\text{NH}_3)_2(\text{en})_2]^{3+}$ (4) $[\text{Cr}(\text{NH}_3)_4(\text{en})]^{3+}$
29. Which among the following will be named as dibromidobis (ethylene diamine) chromium (III) bromide? [AIEEE-2012]
 (1) $[\text{Cr}(\text{en})\text{Br}_2]\text{Br}$ (2) $[\text{Cr}(\text{en})_3]\text{Br}_3$
 (3) $[\text{Cr}(\text{en})_2\text{Br}_2]\text{Br}$ (4) $[\text{Cr}(\text{en})\text{Br}_4]^-$
30. The complex ion $[\text{Pt}(\text{NO}_2)(\text{Py})(\text{NH}_3)(\text{NH}_2\text{OH})]^+$ will give :- [JEE-MAIN-2012, Online]
 (1) 4 isomers (Geometrical)
 (2) 2 isomers (Geometrical)
 (3) 3 isomers (Geometrical)
 (4) 6 isomers (Geometrical)

PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	4	1	3	3	4	3	3	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	2	4	4	1	3	4	4	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	2	3	2	1	1	2	3	3	3

s-Block Elements and Hydrogen & Its Compounds

EXERCISE-I

- The alkali metals which form normal oxide, peroxide as well as super oxides are :-
 (1) Na, Li (2) K, Li
 (3) Li, Cs (4) K, Rb
 - The correct order of degree of hydration of M^+ ions of alkali metals is
 (1) $Li^+ < K^+ < Na^+ < Rb^+ < Cs^+$
 (2) $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$
 (3) $Cs^+ < Rb^+ < K^+ < Na^+ < Li^+$
 (4) $Cs^+ < Rb^+ < Na^+ < K^+ < Li^+$
 - The hydroxide of IInd A metal, which has the lowest value of solubility product (K_{sp}) at normal temperature (25°C) is
 (1) $Ca(OH)_2$ (2) $Mg(OH)_2$
 (3) $Sr(OH)_2$ (4) $Be(OH)_2$
 - $Mg_2C_3 + H_2O \longrightarrow X$ (organic compound). Compound X is
 (1) C_2H_2 (2) CH_4
 (3) propyne (4) ethene
 - The alkaline earth metals, which do not impart any colour to Bunsen flame are
 (1) Be and Mg (2) Mg and Ca
 (3) Be and Ca (4) Be and Ba
 - $Y \xleftarrow{\Delta, 205^\circ C} CaSO_4 \cdot 2H_2O \xrightarrow{\Delta, 120^\circ C} X$.
 X and Y are respectively
 (1) plaster of paris, dead burnt plaster
 (2) dead burnt plaster, plaster of paris
 (3) CaO and plaster of paris
 (4) plaster of paris, mixture of gases
 - The correct order of basic-strength of oxides of alkaline earth metals is
 (1) $BeO > MgO > CaO > SrO$
 (2) $SrO > CaO > MgO > BeO$
 (3) $BeO > CaO > MgO > SrO$
 (4) $SrO > MgO > CaO > BeO$
 - Weakest base among KOH, NaOH, $Ca(OH)_2$ and $Zn(OH)_2$ is
 (1) $Ca(OH)_2$ (2) KOH
 (3) NaOH (4) $Zn(OH)_2$
 - $BeCl_2 + LiAlH_4 \longrightarrow X + LiCl + AlCl_3$
 (1) X is LiH (2) X is BeH_2
 (3) X is $BeCl_2 \cdot 2H_2O$ (4) None
 - A metal which is soluble in both water and liquid NH_3 separately -
 (1) Cr (2) Mn
 (3) Ba (4) Al
 - $MgBr_2$ and MgI_2 are soluble in acetone because of
 (1) Their ionic nature
 (2) Their coordinate nature
 (3) Their metallic nature
 (4) Their covalent nature
 - Which of the following reaction produces hydrogen gas ?
 (1) $Mg + H_2O$ (2) $BaO_2 + HCl$
 (3) $H_2S_2O_8 + H_2O$ (4) $Na_2O_2 + 2HCl$
 - Hydrogen combines with other elements by
 (1) Losing an electron
 (2) Gaining an electron
 (3) Sharing an electron
 (4) Losing, gaining or sharing electron
 - The oxide that gives hydrogen peroxide on the treatment with a dilute acid is
 (1) MnO_2 (2) PbO_2
 (3) Na_2O_2 (4) TiO_2
 - In which of the following reaction hydrogen peroxide is a reducing agent
 (1) $2FeCl_2 + 2HCl + H_2O_2 \longrightarrow 2FeCl_3 + 2H_2O$
 (2) $Cl_2 + H_2O_2 \longrightarrow 2HCl + O_2$
 (3) $2HI + H_2O_2 \longrightarrow 2H_2O + I_2$
 (4) $H_2SO_3 + H_2O_2 \longrightarrow H_2SO_4 + H_2O$
 - When zeolite (Hydrated sodium aluminium silicate) is treated with hard water the sodium ions are exchanged with
 (1) OH^- ions (2) SO_4^{2-} ions
 (3) Ca^{2+} ions (4) H^+ ions
 - Temporary hardness may be removed from water by adding
 (1) $CaCO_3$ (2) $Ca(OH)_2$
 (3) $CaSO_4$ (4) HCl
 - Which of the following can effectively remove all types of hardness of water
 (1) Soap (2) Washing soda
 (3) Slaked lime (4) None of these
 - Temporary unstable hardness of water due to presence of :-
 (1) $CaCl_2, MgSO_4$
 (2) Ca^{+2}, Mg^{+2}
 (3) $K^+, CaCO_3$
 (4) $Ca(HCO_3)_2, Mg(HCO_3)_2$

- 20.** Out of the following metals which will give H_2 on reaction with NaOH :
- I : Zn, II : Mg,
III : Al, IV : Be
(1) I, II, III, IV (2) I, III, IV
(3) II, IV (4) I, III
- 21.** One of the following is an incorrect statement, point it out.
(1) Permanent hardness can be removed by boiling water
(2) Hardness of water effects soap consumption
(3) Temporary hardness is due to bicarbonates of Ca and Mg
(4) Permanent hardness is due to the soluble SO_4^{2-} , Cl^- of Ca and Mg
- 22.** All alkali metal superoxides contain the $[O_2^-]$ ion. They are—
(1) paramagnetic
(2) colored compounds
(3) oxidizing agents
(4) all of these
- 23.** As compared to potassium, sodium has
(1) Lower electronegativity
(2) Higher ionization potential
(3) Larger atomic radius
(4) Lower melting point
- 24.** On passing excess of CO_2 in lime water, its milky appearance disappears because -
(1) Soluble $Ca(OH)_2$ is formed
(2) Soluble $Ca(HCO_3)_2$ is formed
(3) Reaction becomes reversible
(4) Calcium compound evaporated
- 25.** Which of the following alkali metals has the biggest tendency of the half reaction $M_{(g)} \longrightarrow M_{(aq)}^+ + e^-$
(1) Sodium (2) Lithium
(3) Potassium (4) Cesium
- 26.** Which of the following releases 0.2 moles of hydrogen on hydrolysis ?
(1) 0.1 mole of LiH
(2) 0.2 mole of LiH
(3) 0.3 mole of LiH
(4) 0.4 mole of LiH
- 27.** Which of the following statement is not correct ?
(1) LiOH is amphoteric in nature
(2) LiCl is soluble in pyridine
(3) Li_3N is stable while Na_3N doesn't exist even at room temperature
(4) BeO is amphoteric in nature
- 28.** There is loss in weight when mixture of Li_2CO_3 and $Na_2CO_3 \cdot 10H_2O$ is heated strongly. This loss is due to :
(1) Li_2CO_3
(2) $Na_2CO_3 \cdot 10H_2O$
(3) both
(4) none
- 29.** Which of the following statements is incorrect ?
(1) $NaHCO_3$ on heating gives Na_2CO_3
(2) Pure sodium metal dissolves in liquid ammonia to give blue solution
(3) NaOH reacts with glass to give sodium silicate
(4) Aluminium reacts with excess NaOH to give $Al(OH)_3$
- 30.** Which alkali metal on flame test gives red violet colour ?
(1) Li (2) Cs
(3) Na (4) Rb

ANSWER KEY							Exercise-I			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	3	4	3	1	1	2	4	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	1	4	3	2	3	2	2	4	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	4	2	2	2	2	1	3	4	4

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

- The species that do not contain peroxide linkage are - [JEE 1992]
 (1) PbO_2 (2) H_2O_2
 (3) SrO_2 (4) BaO_2
- The following compounds have been arranged in order of their increasing thermal stabilities. Identify the correct order. [JEE 1996]
 $\text{K}_2\text{CO}_3(\text{I})$ $\text{MgCO}_3(\text{II})$ $\text{CaCO}_3(\text{III})$ $\text{BeCO}_3(\text{IV})$
 (1) $\text{I} < \text{II} < \text{III} < \text{IV}$ (2) $\text{IV} < \text{II} < \text{III} < \text{I}$
 (3) $\text{IV} < \text{II} < \text{I} < \text{III}$ (4) $\text{II} < \text{IV} < \text{III} < \text{I}$
- Property of all the alkaline earth metals that increase with their atomic number is - [JEE 1997]
 (1) ionisation energy
 (2) solubility of their hydroxides
 (3) solubility of their sulphate
 (4) electronegativity
- The set representing the correct order of first ionization potential is - [JEE 2001]
 (1) $\text{K} > \text{Na} > \text{Li}$ (2) $\text{Be} > \text{Mg} > \text{Ca}$
 (3) $\text{B} > \text{C} > \text{N}$ (4) $\text{Ge} > \text{Si} > \text{C}$
- A metal M readily forms its sulphate MSO_4 which is water soluble. It forms oxide MO which becomes inert on heating. It forms insoluble hydroxide which is soluble in NaOH. The metal M is :- [AIEEE-2002]
 (1) Mg (2) Ba
 (3) Ca (4) Be
- KO_2 is used in space and submarines because it [AIEEE-2002]
 (1) Absorbs CO_2 and increase O_2 concentration
 (2) Absorbs moisture
 (3) Absorbs CO_2
 (4) Produces ozone
- In curing cement plasters, water is sprinkled from time to time. This helps in :- [AIEEE-2003]
 (1) Hydrating sand and gravel mixed with cement
 (2) Converting sand into silicate
 (3) Developing interlocking needle like crystals of hydrated silicates
 (4) Keeping it cool
- The solubilities of carbonates decreases down the magnesium group due to decrease in :- [AIEEE-2003]
 (1) Inter-ionic attraction
 (2) Entropy of solution formation
 (3) Lattice energy of solids
 (4) Hydration energy of cations
- The substance not likely to contain CaCO_3 is :- [AIEEE-2003]
 (1) Sea shells (2) Dolomite
 (3) A marble statue (4) Calcined gypsum
- One mole of magnesium nitride on reaction with excess of water gives :- [AIEEE-2004]
 (1) Two mole of HNO_3 (2) Two mole of NH_3
 (3) 1 mole of NH_3 (4) 1 mole of HNO_3
- Beryllium and aluminium exhibit many properties which are similar. But the two elements differ in - [AIEEE-2004]
 (1) Exhibiting maximum covalency in compounds
 (2) Forming polymeric hydrides
 (3) Forming covalent halides
 (4) Exhibiting amphoteric nature in their oxides.
- The ionic mobility of alkali metal ions in aqueous solution is maximum for :- [AIEEE-2006]
 (1) Rb^+ (2) Li^+
 (3) Na^+ (4) K^+
- The products obtained on heating LiNO_3 will be :- [AIEEE-2011]
 (1) $\text{LiNO}_2 + \text{O}_2$ (2) $\text{Li}_2\text{O} + \text{NO}_2 + \text{O}_2$
 (3) $\text{Li}_3\text{N} + \text{O}_2$ (4) $\text{Li}_2\text{O} + \text{NO} + \text{O}_2$
- What is the best description of the change that occurs when $\text{Na}_2\text{O}(\text{s})$ is dissolved in water ? [AIEEE-2011]
 (1) Oxidation number of sodium decreases
 (2) Oxide ion accepts sharing in a pair of electrons
 (3) Oxide ion donates a pair of electrons
 (4) Oxidation number of oxygen increases
- Which of the following on thermal-decomposition yields a basic as well as an acidic oxide ? [AIEEE-2012]
 (1) NH_4NO_3 (2) NaNO_3
 (3) KClO_3 (4) CaCO_3
- Very pure hydrogen (99.9%) can be made by which of the following processes ? [AIEEE 2012]
 (1) Reaction of salt like hydrides with water
 (2) Reaction of methane with steam
 (3) Mixing natural hydrocarbons of high molecular weight
 (4) Electrolysis of water
- Based on lattice energy and other considerations, which one of the following alkali metal chloride is expected to have the highest melting point ? [JEE MAIN-2012, Online]
 (1) RbCl (2) LiCl
 (3) KCl (4) NaCl

18. Which one of the following will react most vigorously with water ? [JEE MAIN-2012, Online]
 (1) Li (2) K
 (3) Rb (4) Na
19. A metal M on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour, Y is :- [JEE MAIN-2012, Online]
 (1) NH_3 (2) MgO
 (3) Mg_3N_2 (4) $Mg(NO_3)_2$
20. The correct statement for the molecule, CsI_3 , is : [JEE(Main)-2014]
 (1) it contains Cs^{3+} and I^- ions
 (2) it contains Cs^+ , I^- and lattice I_2 molecule
 (3) it is a covalent molecule
 (4) it contains Cs^+ and I_3^- ions
21. Which of the following statements about Na_2O_2 is not correct ? [JEE MAIN-2014, Online]
 (1) Na_2O_2 oxidises Cr^{3+} to CrO_4^{2-} in acid medium
 (2) It is diamagnetic in nature
 (3) It is the super oxide of sodium
 (4) It is a derivative of H_2O_2
22. Amongst $LiCl$, $RbCl$, $BeCl_2$ and $MgCl_2$ the compounds with the greatest and the least ionic character, respectively are : [JEE MAIN-2014, Online]
 (1) $RbCl$ and $MgCl_2$ (2) $LiCl$ and $RbCl$
 (3) $MgCl_2$ and $BeCl_2$ (4) $RbCl$ and $BeCl_2$
23. From the following statements regarding H_2O_2 , choose the incorrect statement : [JEE(Main) 2015]
 (1) It has to be stored in plastic or wax lined glass bottles in dark
 (2) It has to be kept away from dust
 (3) It can act only as an oxidizing agent
 (4) It decomposes on exposure to light
24. The correct order of thermal stability of hydroxides is : [JEE(Main)Online-2015]
 (1) $Ba(OH)_2 < Sr(OH)_2 < Ca(OH)_2 < Mg(OH)_2$
 (2) $Mg(OH)_2 < Sr(OH)_2 < Ca(OH)_2 < Ba(OH)_2$
 (3) $Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$
 (4) $Ba(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Mg(OH)_2$
25. Which of the alkaline earth metal halides given below is essentially covalent in nature :- [JEE(Main)Online-2015]
 (1) $SrCl_2$ (2) $CaCl_2$
 (3) $BeCl_2$ (4) $MgCl_2$
26. Which one of the following alkaline earth metal sulphates has its hydration enthalpy greater than its lattice enthalpy ? [JEE(Main)-2015]
 (1) $BaSO_4$ (2) $SrSO_4$
 (3) $CaSO_4$ (4) $BeSO_4$
27. The commercial name for calcium oxide is : [JEE(Main)-2016]
 (1) Quick lime (2) Milk of lime
 (3) Limestone (4) Slaked lime
28. The correct order of the solubility of alkaline-earth metal sulphates in water is : [JEE(Main)-2016]
 (1) $Mg < Sr < Ca < Ba$
 (2) $Mg < Ca < Sr < Ba$
 (3) $Mg > Ca > Sr > Ba$
 (4) $Mg > Sr > Ca > Ba$
29. The main oxides formed on combustion of Li, Na and K in excess of air are respectively : [JEE(Main)-2016]
 (1) Li_2O , Na_2O_2 and KO_2
 (2) Li_2O , Na_2O and KO_2
 (3) LiO_2 , Na_2O_2 and K_2O
 (4) Li_2O_2 , Na_2O_2 and KO_2
30. In KO_2 , the nature of oxygen species and the oxidation state of oxygen atom are, respectively [JEE(Main)ONLINE-2018]
 (1) Superoxide and $-1/2$
 (2) Oxide and -2
 (3) Peroxide and $-1/2$
 (4) Superoxide and -1

PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	2	2	4	1	3	4	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	4	3	4	4	4	3	3	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	4	3	3	3	4	1	3	1	1

METALLURGY

EXERCISE-I

- Aluminium is obtained from Al_2O_3 by this method
(1) Thermal reduction.
(2) Hydro metallurgical method.
(3) Electrolytic reduction.
(4) Reduction by iron.
- Zinc blende on roasting in air gives :-
(1) Zinc carbonate (2) SO_2 and ZnO
(3) ZnS and ZnSO_4 (4) CO_2 and ZnO
- The oxide cannot be reduced by coke
(1) Cu_2O , ZnO (2) Fe_2O_3 , ZnO
(3) CaO , K_2O (4) PbO , Fe_3O_4
- Which is not a basic flux :-
(1) Silica (2) Lime stone
(3) Calcite (4) Quick lime
- Iron pyrites ore is concentrated by:-
(1) Froth floatation (2) Electrolysis
(3) Roasting (4) Magnetic separation
- In Goldschmidt thermite process, reducing agent is:-
(1) Fe (2) Na
(3) Ca (4) Al
- Liquation process is used for refining:-
(1) Bismuth (2) Lead
(3) Tin (4) All
- Autoreduction process is used in the extraction of:-
(1) Cu & Pb (2) Zn & Hg
(3) Cu & Al (4) Fe & Pb
- In the electrolytic refining of copper, Ag and Au are found:-
(1) On cathode (2) On anode
(3) In the anodic mud (4) In the cathodic mud
- Which of the following metals can not be extracted by carbon reduction process :-
(1) Pb (2) Al
(3) Sn (4) Zn
- The maximum temperature obtained in the....region of the blast furnace used in extraction of iron:-
(1) Reduction (2) Combustion
(3) Fusion (4) Slag formation
- Which of the following process involves smelting
(1) $2 \text{PbS} + 3 \text{O}_2 \rightarrow 2 \text{PbO} + 2 \text{SO}_2 \uparrow$
(2) $\text{Al}_2\text{O}_3 \cdot 2 \text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 2 \text{H}_2\text{O}$
(3) $\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow 2 \text{Fe} + 2 \text{CO}_2$
(4) $\text{Cr}_2\text{O}_3 + 2 \text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2 \text{Cr} + \text{Heat}$
- In the extraction of copper from pyrites, iron is removed as:-
(1) FeSO_4 (2) FeSiO_3
(3) Fe_3O_4 (4) Fe_2O_3
- Which one of the following metals can not be extracted by using Al as a reducing agent :-
(1) Na from Na_2O (2) Cr from Cr_2O_3
(3) Mn from MnO_2 (4) V from V_2O_5
- Column - I** **Column - II**
(A) Metal which occur in the native state in nature is (P) Hg
(B) The oxides of metal that can be commercially reduced by Aluminothermite reduction process is (Q) Ti
(C) van Arkel method is used for preparing ultrapure metal of (R) Cr
(D) Auto reduction process is employed for the sulphide ore of (S) Ag
(1) A-S, B-R, C-Q, D-P (2) A-R, B-S, C-Q, D-P
(3) A-P, B-S, C-Q, D-R (4) A-Q, B-R, C-S, D-P
- Which of the following pairs of metals is purified by Van Arkel method ?
(1) Ga and In (2) Zr and Ti
(3) Ag and Au (4) Ni and Fe
- The following reactions take place in the blast furnace in the preparation of impure iron. Identify the reaction pertaining to the formation of the slag:-
(1) $2 \text{C(s)} + \text{O}_2(\text{g}) \rightarrow 2 \text{CO(g)}$
(2) $\text{Fe}_2\text{O}_3(\text{s}) + 3 \text{CO(g)} \rightarrow 2 \text{Fe(l)} + 3 \text{CO}_2(\text{g})$
(3) $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO(s)} + \text{CO}_2(\text{g})$
(4) $\text{CaO(s)} + \text{SiO}_2(\text{s}) \rightarrow \text{CaSiO}_3(\text{s})$
- Aluminium is extracted from alumina (Al_2O_3) by electrolysis of a molten mixture of:
(1) $\text{Al}_2\text{O}_3 + \text{Na}_3\text{AlF}_6 + \text{CaF}_2$
(2) $\text{Al}_2\text{O}_3 + \text{KF} + \text{Na}_3\text{AlF}_6$
(3) $\text{Al}_2\text{O}_3 + \text{HF} + \text{NaAlF}_4$
(4) $\text{Al}_2\text{O}_3 + \text{CaF}_2 + \text{NaAlF}_4$
- Which one of the following is not a method of concentration of ore?
(1) gravity separation
(2) froth floating process
(3) electromagnetic separation
(4) smelting

- 20.** Chemical leaching is useful in the concentration of:
 (1) copper pyrites (2) bauxite
 (3) galena (4) cassiterite
- 21.** During initial treatment, preferential wetting of ore by oil and gangue by water takes place in
 (1) Levigation (gravity separation)
 (2) Froth floatation
 (3) Leaching
 (4) Bessemerisation
- 22.** The benefactions of the sulphide ores is usually done by
 (1) Electrolysis
 (2) Smelting process
 (3) Metal displacement method
 (4) Froth floatation method
- 23.** Calcination is the process of heating the ore:
 (1) in inert gas
 (2) in the presence of air
 (3) in the absence of air
 (4) in the presence of CaO and MgO
- 24.** When roasting is carried out :
 (i) Sulphide ore are convert to oxide and sulphate
 (ii) remove water of hydration
 (iii) melt the ore
 (iv) remove arsenic and sulphur impurities
 Of these statements:
 (1) (i), (ii) and (iii) are correct
 (2) (i) and (iv) are correct
 (3) (i), (ii) and (iv) are correct
 (4) (ii), (iii) and (iv) are correct
- 25.** In the alumino thermite process, Al acts as
 (1) An oxidising agent
 (2) A flux
 (3) A reducing agent
 (4) A solder
- 26.** Which of the following employ(s) thermal decomposition of volatile iodide compounds?
 (1) Thermite process
 (2) Hall's process
 (3) Van-Arkel's process
 (4) Mond's process
- 27.** Refining of silver is done by:
 (1) liquation (2) poling
 (3) cupellation (4) van Arkel method
- 28.** Iron obtained from blast furnace is:
 (1) wrought iron (2) cast iron
 (3) pig iron (4) steel
- 29.** Which of the following term is not related to Al-extraction
 (1) Serpek's process
 (2) Hall-Heroult process
 (3) Thermite process
 (4) Hoop's process
- 30.** Silica is added to roasted copper ores during extraction in order to remove
 (1) cuprous sulphide
 (2) ferrous oxide
 (3) ferrous sulphide
 (4) cuprous oxide

				ANSWER KEY			Exercise-I			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	3	1	1	4	4	1	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	2	1	1	2	4	1	4	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	4	3	3	3	3	3	3	3	2

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

1. In the commercial electrochemical process for aluminium extraction, the electrolyte used as:

[JEE-1999]

- (1) $\text{Al}(\text{OH})_3$ in NaOH solution
- (2) an aqueous solution of $\text{Al}_2(\text{SO}_4)_3$
- (3) a molten mixture of Al_2O_3 and Na_3AlF_6
- (4) a molten mixture of $\text{AlO}(\text{OH})$ and $\text{Al}(\text{OH})_3$

2. Electrolytic reduction of alumina to aluminium by Hall-Heroult process is carried out:

[IIT-2000]

- (1) in the presence of NaCl
- (2) in the presence of fluorite
- (3) in the presence of cryolite which forms a melt with lower melting temperature
- (4) in the presence of cryolite which forms a melt with higher melting temperature

3. The chemical composition of "slag" formed during the smelting process in the extraction of copper is:

[IIT-2001]

- (1) $\text{Cu}_2\text{O} + \text{FeS}$
- (2) FeSiO_3
- (3) CuFeS_2
- (4) $\text{Cu}_2\text{S} + \text{FeO}$

4. Aluminium is extracted by the electrolysis of :-

[AIEEE-2002]

- (1) Bauxite
- (2) Alumina
- (3) Alumina mixed with molten cryolite
- (4) Molten cryolite

5. Pyrolusite is a/an :-

[AIEEE-2002]

- (1) Oxide ore
- (2) Sulphide ore
- (3) Carbide ore
- (4) Not an ore

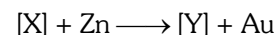
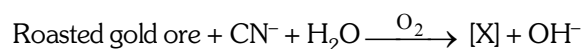
6. Which of the following processes is used in extractive metallurgy of magnesium?

[IIT-2002]

- (1) Fused salt electrolysis
- (2) Self reduction
- (3) Aqueous solution electrolysis
- (4) Thermite reduction

7. In the process of extraction of gold,

[IIT-2003]



Identify the complexes [X] and [Y] :

- (1) $\text{X} = [\text{Au}(\text{CN})_2]^-$, $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$
- (2) $\text{X} = [\text{Au}(\text{CN})_4]^{3-}$, $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$
- (3) $\text{X} = [\text{Au}(\text{CN})_2]^-$, $\text{Y} = [\text{Zn}(\text{CN})_6]^{4-}$
- (4) $\text{X} = [\text{Au}(\text{CN})_4]^-$, $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$

8. Which one of the following ores is best concentrated by froth-flotation method :

[AIEEE-2004]

- (1) Galena
- (2) Cassiterite
- (3) Magnetite
- (4) Malachite

9. The methods chiefly used for the extraction of lead and tin from their ores are respectively:

[IIT-2004]

- (1) self reduction and carbon reduction
- (2) self reduction and electrolytic reduction
- (3) carbon reduction and self reduction
- (4) cyanide process and carbon reduction

10. Which ore contains both iron and copper ?

[IIT-2004]

- (1) Cuprite
- (2) Chalcocite
- (3) Chalcopyrite
- (4) Malachite

11. Extraction for zinc from zinc blende is achieved by:

[IIT-2007]

- (1) electrolytic reduction
- (2) roasting followed by reduction with carbon
- (3) roasting followed by reduction with another metal
- (4) roasting followed by self-reduction

12. Native silver metal forms a water soluble complex with a dilute aqueous solution of NaCN in the presence of

[IIT-2008]

- (1) nitrogen
- (2) oxygen
- (3) carbon dioxide
- (4) argon

Paragraph for questions 13 and 14

Copper is the most noble of the first row transition metals and occurs in small deposits in several countries. Ores of copper include chalcantite ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$), atacamite ($\text{Cu}_2\text{Cl}(\text{OH})_3$), cuprite (Cu_2O), copper glance (Cu_2S) and malachite ($\text{Cu}_2(\text{OH})_2\text{CO}_3$). However, 80% of the world copper production comes from the ore chalcopyrite (CuFeS_2). The extraction of copper from chalcopyrite involves partial roasting, removal of iron and self-reduction. [IIT-2010]

13. Iron is removed from chalcopyrite as :-

- (1) FeO (2) FeS
(3) Fe_2O_3 (4) FeSiO_3

14. In self-reduction, the reducing species is :-

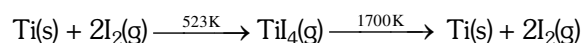
- (1) S (2) O^{2-}
(3) S^{2-} (4) SO_2

15. Oxidation states of the metal in the minerals haematite and magnetite, respectively, are

[JEE-2011]

- (1) II, III in haematite and III in magnetite
(2) II, III in haematite and II in magnetite
(3) II in haematite and II, III in magnetite
(4) III in haematite and II, III in magnetite

16. Which method of purification is represented by the following equation : [AIEEE-2012]



- (1) Van Arkel (2) Zone refining
(3) Cupellation (4) Poling

17. In the cyanide extraction process of silver from argentite ore, the oxidizing and reducing agents used are : [JEE-2012]

- (1) O_2 and CO respectively.
(2) O_2 and Zn dust respectively.
(3) HNO_3 and Zn dust respectively.
(4) HNO_3 and CO respectively.

18. Sulfide ores are common for the metals -

[JEE-2013]

- (1) Ag, Cu and Pb (2) Ag, Cu and Sn
(3) Ag, Mg and Pb (4) Al, Cu and Pb

19. The carbon-based reduction method is **NOT** used for the extraction of [JEE-2013]

- (1) tin from SnO_2
(2) Iron from Fe_2O_3
(3) aluminium from Al_2O_3
(4) magnesium from $\text{MgCO}_3 \cdot \text{CaCO}_3$

PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	2	3	1	1	1	1	1	3
Que.	11	12	13	14	15	16	17	18	19	
Ans.	2	2	4	3	4	1	2	1	3,4	